



An ISO 9001 : 2008 Certified Institute

India's Pioneer Institute of
MATHEMATICS

NET/JRF | GATE | IIT-JAM | UPSC | M.Sc. Ent. (DU)

*A High Quality Study Material
for
Higher Level Exam for U.G. & P.G. Students*

Statistics

▲ Ph. : 011-26537527, 9999183434, 9899161734, 8588844789 ▲

STATISTICS

Sandeep MALIK

98020-63796

Chapter 1: Probability

1.1	Outcomes and Events	1
1.2	Probability Functions/Measures	1
1.2.1	Properties of $P[\cdot]$	1
1.3	Conditional Probability and Independence	2

Chapter 2: Random Variables

2.1	Random Variables and Cumulative Distribution Functions	4
2.2	Density Functions	5
2.3	Expectations and Moments	5
2.4	Expectation of a Function of a Random Variable	6
2.5	Two Important Inequality Results	7
2.6	Moments and Moment Generating Functions	7
2.7	Other Distribution Summaries	8

Chapter 3: Special Univariate Distributions

3.1	Discrete Distributions	9
3.1.1	Degenerate Distribution	9
3.1.2	Two Point Distribution	9
3.1.3	Uniform Distribution on n - Points	9
3.1.4	Binomial Distribution	10
3.1.5	Negative Binomial Distribution (Pascal or Waiting time distribution)	10
3.1.6	Hyper-Geometric Distribution	11
3.1.7	Poisson Distribution	12
3.1.8	Multi-Nomial Distribution: (Generalized Binomial Distribution)	13
3.2	Continuous Distributions	14
3.2.1	Uniform Distribution	14
3.2.2	Gamma Distribution	14
3.2.3	Beta Distribution	15
3.2.4	Normal Distribution (Gaussian Law)	16
3.2.5	Cauchy Distribution	17

Chapter 4: Joint and Conditional Distributions

4.1	Joint Distributions	18
4.2	Special Multivariate Distributions	18
4.2.1	Multinomial Distribution	19
4.2.2	Bi-Variate Normal Distribution	19
4.3	Conditional Distributions and Densities	20
4.4	Conditional Expectation	20
4.5	Conditional Expectations of Functions of Random Variables	21
4.6	Independence of Random Variables	21
4.7	Covariance and Correlation	21
4.8	Transformation of Random Variables: $Y = g(X)$	22
4.9	Moment-Generating-Function Technique	23

Chapter 5: Inference

5.1	Sample Statistics	24
5.2	Sampling Distributions	24
5.3	Point Estimation	27
5.4	Interval Estimation	29

Chapter 6: Maximum Likelihood Estimation

6.1	Likelihood Function and ML Estimator	34
6.4	Properties of MLEs	35

Chapter 7: Hypothesis Testing

7.1	Introduction	36
7.2	Simple Hypothesis Tests	37
7.3	Simple Null, Composite Alternative	40
7.4	Composite Hypothesis Tests	41

Chapter 8: Elements of Bayesian Inference

8.1	Introduction	44
8.2	Parameter Estimation	45
8.3	Conjugate Prior Distributions	47

Chapter 9: Markov Chains

9.1	Discrete Time Markov Chain	49
-----	----------------------------------	----

Chapter 10: Miscellaneous

10.1	Reliability Analysis	55
10.2	Quadratic Forms	57
	Assignment Sheet-1	59
	Assignment Sheet-2	63
	Assignment Sheet-3	71
	Assignment Sheet-4	75

CHAPTER 1

PROBABILITY

1.1. Outcomes and Events

We consider experiments, which comprise: a collection of distinguishable outcomes, which are termed elementary events, and typically denoted by Ω and a collection of sets of possible outcomes to which we might wish to assign probabilities, $\mathcal{A}$ the event.

In order to obtain a sensible theory of probability, we require that our collection of events $\mathcal{A}$ is an algebra over, i.e. it must possess the following properties

- (i.) $\Omega \in \mathcal{A}$
- (ii.) If A in $\mathcal{A}$, then $\bar{A} \in \mathcal{A}$
- (iii.) If A_1 and $A_2 \in \mathcal{A}$, then $A_1 \cup A_2 \in \mathcal{A}$.

In the case of finite Ω , we might note that the collection of all subsets of Ω necessarily satisfies the above properties and by using this default choice of algebra, we can assign probabilities to any possible combination of elementary events.

Proposition 1.1: If $\mathcal{A}$ is an algebra, then $\phi \in \mathcal{A}$.

Proposition 1.2: If A_1 and $A_2 \in \mathcal{A}$, then $A_1 \cap A_2 \in \mathcal{A}$ for any algebra $\mathcal{A}$.

Proposition 1.3: If $\mathcal{A}$ is an algebra and $A_1, A_2, \dots, A_n \in \mathcal{A}$, then $\bigcap_{i=1}^n A_i \in \mathcal{A}$.

1.2. Probability Functions/Measures

Let Ω denote the sample space and $\mathcal{A}$ denote a collection of events assumed to be a σ -algebra.

Definition 1.1: (Probability Function): A probability function $P[\cdot]$ is a set function with domain $\mathcal{A}$ (a σ -algebra of events) and range $[0, 1]$, i.e., $P: \mathcal{A} \rightarrow [0, 1]$, which satisfies the following axioms

- (i.) $P[A] \geq 0$ for every $A \in \mathcal{A}$
- (ii.) $P[\Omega] = 1$
- (iii.) If $A_1, A_2, \dots$ is a sequence of mutually exclusive events (i.e. $A_i \cap A_j = \phi$

for any $i \neq j$) in $\mathcal{A}$ and if $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$, then $P\left[\bigcup_{i=1}^{\infty} A_i\right] = \sum_{i=1}^{\infty} P[A_i]$

1.2.1. Properties of $P[\cdot]$

A remarkably rich theory emerges from these three axioms (together, of course, with those of set theory). Indeed, all formal probability follows as a logical consequence of these axioms. Some of the most important simple results are summarised here. Throughout this section, assume that Ω is our collection of possible outcomes $\mathcal{A}$ is a σ -algebra over Ω and $P[\cdot]$ is an associated probability distribution.

Many of these results simply demonstrate that things which we would intuitively want to be true of probabilities do, indeed, arise as logical consequences of this simple axiomatic framework.

Proposition 1.4: $P[\phi] = 0$.

Proposition 1.5: If $A_1, A_2, \dots, A_n$ are pairwise disjoint elements of $\mathcal{A}$, corresponding to mutually exclusive outcomes in our experiment, then

$$P[A_1 \cup A_2 \cup \dots \cup A_n] = \sum_{i=1}^n P[A_i]$$

Proposition 1.6: If $A \in \mathcal{A}$ then

$$P[\bar{A}] = 1 - P[A]$$

Proposition 1.7: For any two events $A, B \in \mathcal{A}$

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

Proposition 1.8: If $A, B \in \mathcal{A}$ and $A \subset B$, then

$$P[A] \leq P[B]$$

Proposition 1.9 (Boole's Inequality): If $A_1, \dots, A_n \in \mathcal{A}$, then

$$P[A_1 \cup A_2 \cup \dots \cup A_n] \leq P[A_1] + P[A_2] + \dots + P[A_n].$$

Definition 1.2: (Probability Space): A probability space is the triple $(\Omega, \mathcal{A}, P[\cdot])$, where Ω is a sample space, $\mathcal{A}$ is a σ -algebra over Ω , and $P[\cdot]$ is a probability function with domain $\mathcal{A}$.

1.3. Conditional Probability and Independence

Sometimes it's possible to observe that one event has occurred. In this situation, we wish to have a model for the behaviour of the probability that other events compatible with B . Conditional probability is the appropriate language.

Definition 1.3: (Conditional Probability): Let A and B be events in $\mathcal{A}$ of the given probability space $(\Omega, \mathcal{A}, P[\cdot])$. The conditional probability of event A given event B , denoted by $P[A|B]$, is defined as

$$P[A|B] = \frac{P[A \cap B]}{P[B]} \text{ if } P[B] > 0,$$

and is left undefined when $P[B] = 0$.

Exercise 1.3.1: Consider the experiment of tossing two coins, $\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$, and assume that each point is equally likely. Find

- The probability of two heads given a head on the first coin.
- The probability of two heads given at least one head.

Theorem 1.1: (Law of Total Probability): For a given probability space $(\Omega, \mathcal{A}, P[\cdot])$, if $B_1, \dots, B_n$ is a collection of mutually disjoint events in $\mathcal{A}$ satisfying

$$\Omega = \bigcup_{j=1}^n B_j,$$

i.e. $B_1, \dots, B_n$ partition Ω and $P[B_j] > 0$, $j = 1, \dots, n$, then for every $A \in \mathcal{A}$,

$$P[A] = \sum_{j=1}^n P[A \cap B_j].$$

Conditional probability has a number of useful properties. The following elementary result is surprisingly important and has some far-reaching consequences.

Theorem 1.2 (Bayes' Formula): For a given probability space $(\Omega, \mathcal{A}, P[\cdot])$, if $A, B \in \mathcal{A}$ are such that $P[A] > 0$, $P[B] > 0$, then:

$$P[A|B] = \frac{P[B|A]P[A]}{P[B]}.$$

Theorem 1.3 (Partition Formula): If $B_1, \dots, B_n \in \mathcal{A}$ partition Ω , then for any $A \in \mathcal{A}$:

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i).$$

Theorem 1.4 (Multiplication Rule): For a given probability space $(\Omega, \mathcal{A}, P[\cdot])$, let $A_1, \dots, A_n$ be events belonging to $\mathcal{A}$ for which $P[A_1, \dots, A_{n-1}] > 0$, then

$$P[A_1, A_2, \dots, A_n] = P[A_2|A_1] \dots P[A_n|A_1, \dots, A_{n-1}].$$

Definition 1.4 (Independent Events): For a given probability space $(\Omega, \mathcal{A}, P[\cdot])$, let A and B be two events in $\mathcal{A}$. Events A and B are defined to be independent iff one of the following conditions is satisfied.

- (i.) $P[A \cap B] = P[A]P[B]$
- (ii.) $P[A|B] = P[A]$ if $P[B] > 0$
- (iii.) $P[B|A] = P[B]$ if $P[A] > 0$.

Exercise 1.3.2: Consider the experiment of rolling two dice. Let $A = \{\text{total is odd}\}$, $B = \{6 \text{ on the first die}\}$, $C = \{\text{total is seven}\}$.

- (i.) Are A and B independent?
- (ii.) Are A and C independent?
- (iii.) Are B and C independent?

Definition 1.5 (Independence of Several Events): For a given probability space $(\Omega, \mathcal{A}, P[\cdot])$, let $A_1, \dots, A_n$ be events in $\mathcal{A}$. Events $A_1, \dots, A_n$ are defined to be independent iff

- (i) $P[A_i \cap A_j] = P[A_i]P[A_j]$, $i \neq j$
- (ii) $P[A_i \cap A_j \cap A_k] = P[A_i]P[A_j]P[A_k]$, $i \neq j, j \neq k, i \neq k$

...

$$n. P\left[\bigcap_{i=1}^n A_i\right] = \prod_{i=1}^n P[A_i].$$

Put Your Own Notes

CHAPTER 2

RANDOM VARIABLES

2.1 Random Variables and Cumulative Distribution Functions

We considered random events in the previous chapter: experimental outcomes which either do or do not occur. In general we cannot predict whether or not a random event will or will not occur before we observe the outcome of the associated experiment — although if we know enough about the experiment we may be able to make good probabilistic predictions. The natural generalization of a random event is a random variable: an object which can take values in the set of real numbers (rather than simply happening or not happening) for which the precise value which it takes is not known before the experiment is observed.

The following definition may seem a little surprising if you've seen probability only outside of measure-theoretic settings in the past. In particular, random variables are deterministic functions: neither random nor variable in themselves. This definition is rather convenient; all randomness stems from the underlying probability space and it is clear that random variables and random events are closely related. This definition also makes it straightforward to define multiple random variables related to a single experiment and to investigate and model the relationships between them.

Definition 2.1 (Random Variable): Given a probability space $(\Omega, \mathcal{A}, \mathbf{P}[\cdot])$, a **random variable**, X , is a function with domain Ω and co-domain $\mathbb{R}$ (the real line) (i.e., $X : \Omega \rightarrow \mathbb{R}$).

Example 2.1: Roll 2 dice $\Omega = \{(i, j); i, j = 1, \dots, 6\}$. Several random variables can be defined, for example $X(i, j) = i + j$, also $Y((i, j)) = |i - j|$. Both, X and Y are random variables. X can take values $2, 3, \dots, 12$ and Y can take values $0, 1, \dots, 5$.

Definition 2.2 (Distribution Function): The **distribution function** of a random variable X , denoted by $F_X(\cdot)$ is defined to be the function $F_X : \mathbb{R} \rightarrow [0, 1]$ which assigns

$$F_X(x) = \mathbf{P}[X \leq x] = \mathbf{P}[\{\omega : X(\omega) \leq x\}]$$

for every $x \in \mathbb{R}$.

Properties of $F_X(\cdot)$

- (i) $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow +\infty} F_X(x) = 1$
- (ii) $F_X(a) \leq F_X(b)$ for $a < b$ (monotonic and non-decreasing)
- (iii) $F_X(\cdot)$ is continuous from the right

$$\lim_{h \downarrow 0} F_X(x+h) = F_X(x)$$

Any function $F(\cdot)$ with $F : \mathbb{R} \rightarrow [0, 1]$ satisfying the above properties is a distribution function for some random variable.

2.2 Density Functions

For two distinct classes of random variables, the distribution of values can be described more simply by using density functions. These classes are termed 'discrete' and 'continuous'.

Definition 2.3 (Discrete Random Variable): A random variable X is **discrete** if the range of X is countable (i.e. it is finite or isomorphic to the natural numbers).

Definition 2.4 (Discrete Density Function): If X is a discrete random variable with distinct values $x_1, x_2, \dots, x_n, \dots$, then the discrete density function of X is defined by

$$f_X(x) = \begin{cases} P[X = x_j] & \text{if } x = x_j, j = 1, 2, \dots, n, \dots \\ 0 & \text{if } x \neq x_j \end{cases}$$

Exercise 2.2.1: Consider the experiment of tossing two dice. Let $X = \{\text{total of upturned faces}\}$ and $Y = \{\text{absolute difference of upturned faces}\}$.

- Give the probability function f_X and sketch it.
- Give f_Y .

Definition 2.5: Any function $F: \mathbb{R} \rightarrow [0, 1]$ is defined to be a discrete density function if for some countable set $x_1, x_2, \dots, x_n, \dots$

- $f(x_j) \geq 0 \quad j = 1, 2, \dots$
- $f(x) = 0$ for $x \neq x_j; j = 1, 2, \dots$
- $\sum_j f(x_j) = 1$ where summation is over $x_1, x_2, \dots, x_n, \dots$

Definition 2.6 (Continuous Random Variable): A random variable X is called continuous if there exists a function $F_X(\cdot)$ such that

$$F_X(x) = \int_{-\infty}^x f_X(u) du \quad \text{for every } x \in \mathbb{R}.$$

Definition 2.7 (Probability Density Function of a Continuous Random Variable): If X is a continuous random variable, the function f_X in

$$F_X(x) = \int_{-\infty}^x f_X(u) du \quad \text{is called the probability density function of } X.$$

Definition 2.8: A function $f: \mathbb{R} \rightarrow [0, \infty)$ is a probability density function iff

- $f(x) \geq 0 \quad \forall x$
- $\int_{-\infty}^{+\infty} f(x) dx = 1.$

2.3 Expectations and Moments

Definition 2.9 (Expectation, Mean): Let X be a random variable. The mean of X , denoted by μ_X or $E[X]$, is defined by

- $E[X] = \sum_j x_j f_X(x_j)$ if X is discrete with mass points $x_1, x_2, \dots, x_j, \dots$
- $E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$ if X is continuous with density $f_X(x)$.

Intuitively, $E[X]$ is the centre of gravity of the unit mass that is specified by the density function.

Put Your Own Notes

Exercise 2.3.1: Consider the experiment of rolling two dice. Let X denote the total of two dice and Y their absolute difference. Compute $E[X]$ and $E[Y]$.

Exercise 2.3.2: Let X be a continuous random variable with density

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } 0 \leq x < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Compute $E[X]$ and $F_X(x)$.

Definition 2.10 (Variance): Let X be a random variable, and let $\mu_X = E[X]$. The variance of X , denoted by σ_X^2 or $\text{Var}[X]$ is defined by

- (i) $\text{Var}[X] = \sum_j (x_j - \mu_X)^2 f_X(x_j)$ if X is discrete with mass points $x_1, x_2, \dots, x_j, \dots$
- (ii) $\text{Var}[X] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx$ for continuous X with density $f_X(x)$.

Variance is a measure of spread or dispersion. If the values of a random variable X tend to be far from their mean, the variance of X will be larger than the variance of a comparable random variable Y whose values are typically nearer to the mean.

Definition 2.11 (Standard Deviation): If X is a random variable, the standard deviation of X , denoted by σ_X , is defined as $+\sqrt{\text{Var}[X]}$.

2.4 Expectation of a Function of a Random Variable

Definition 2.12 (Expectation of a Function of a Random Variable): Let X be a random variable and $g(\cdot)$ a function $g(\cdot): \mathbb{R} \rightarrow \mathbb{R}$. The expectation or expected value of the function $g(\cdot)$ of the random variable X , denoted by $E[g(X)]$ is defined by

- (i) $E[g(X)] = \sum_j g(x_j) f_X(x_j)$ if X is discrete with mass points $x_1, x_2, \dots, x_j, \dots$ and provided the series is absolutely convergent.
- (ii) $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$, for continuous X with density $f_X(x)$, provided the integral exists.

If $g(X) = X$, then $E[g(X)] = E[X] \equiv \mu_X$. If $g(X) = (X - \mu_X)^2$, then $E[g(X)] = \sigma_X^2$.

Proposition 2.1 (Properties of Expectations): Expectations have a number of useful properties which are very often useful:

- (i) $E[c] = c$ for a constant c
- (ii) $E[cg(X)] = cE[g(X)]$ for a constant c
- (iii) $E[c_1g_1(X) + c_2g_2(X)] = c_1E[g_1(X)] + c_2E[g_2(X)]$
- (iv) $E[g_1(X)] \leq E[g_2(X)]$ if $g_1(x) \leq g_2(x) \quad \forall x$

Proposition 2.2 (Variance in Terms of Expectations): If X is a random variable, then

$$\text{Var}[X] = E[(X - E(X))^2] = E[X^2] - (E[X])^2$$

2.5 Two Important Inequality Results

Theorem 2.1 (Chebyshev's Inequality): Let X be a random variable and $g(\cdot)$ a non-negative function, then

$$P[g(X) \geq k] \leq \frac{E[g(X)]}{k} \quad \forall k > 0.$$

Corollary 2.1: If X is a random variable with finite variance, σ_X^2 , then:

$$P[|X - \mu_X| \geq r\sigma_X] = P[(X - \mu_X)^2 \geq r^2\sigma_X^2] \leq \frac{1}{r^2}.$$

Note, that the last statement can also be written as

$$P[|X - \mu_X| \leq r\sigma_X] \geq 1 - \frac{1}{r^2}.$$

Thus the probability that X falls within $r\sigma$ units of μ_X is greater than or equal to $1 - \frac{1}{r^2}$. For $r = 2$ one gets

$$P[\mu_X - 2\sigma_X < X < \mu_X + 2\sigma_X] \geq \frac{3}{4}$$

or, for each random variable X having a finite variance, at least 3/4 of the mass of X falls within two standard deviations of its mean. Chebyshev's inequality gives a bound which does not depend on the distribution of X .

Theorem 2.2 (Jensen's Inequality): For a RV X with mean $E[X]$ and $g(\cdot)$ a convex continuous function,

$$E[g(X)] \geq g(E[X]).$$

Note, a convex function, g , is any function which lies beneath any of its chords. That is, given points a and b , with $a < b$, for any $\lambda \in [0, 1]$:

$$g(a + \lambda(b - a)) = g((1 - \lambda)a + \lambda b) \leq (1 - \lambda)g(a) + \lambda g(b).$$

2.6 Moments and Moment Generating Functions

Definition 2.13 (Moments and Central Moments): For a RV X , the r^{th} moment of X is given by

$$\mu'_r = E[X^r]$$

If the expectation exists. For a RV X , the r^{th} central moment of X (the r^{th} moment about μ'_1) is given by

$$\mu_r = E[(X - \mu'_1)^r] = E\left[\sum_{i=1}^r \binom{r}{i} (\mu'_1)^{r-i} X^i\right] = \sum_{i=1}^r \binom{r}{i} (\mu'_1)^{r-i} E[X^i]$$

The First Four Central Moments.

Zero $\mu_1 = E[(X - \mu_X)] = 0$

Variance $\mu_2 = E[(X - \mu_X)^2]$

Skewness $\frac{\mu_3}{\sigma_X^3} = \frac{E[(X - \mu_X)^3]}{\sigma_X^3}$

Kurtosis $\frac{\mu_4}{\sigma_X^4} = \frac{E[(X - \mu_X)^4]}{\sigma_X^4}$

Note that all odd central moments of X about μ_X are 0 if the density function is symmetrical about μ_X . The skewness is used to indicate whether a density is skewed to the left (value for skewness is negative) or skewed to the right (value for skewness is positive). The kurtosis is sometimes used to indicate that a density is more peaked (and heavier tailed) around its centre than the normal density.

The moment generating function gives us a function from which all the moments can be reconstructed via differentiation.

Definition 2.14 (Moment Generating Function (MGF)): Let X be a RV with density $f_X(\cdot)$. We may define its **moment generating function**, $m(t)$, as:

Discrete

$$m(t) = E[e^{tX}] = \sum_x e^{tx} f_X(x)$$

Continuous

$$m(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$$

The MGF has the property that

$$\left. \frac{d^r m(t)}{dt^r} \right|_{t=0} = \mu'_r$$

Theorem 2.3 (Equality of Distributions): Let X and Y be two random variables with densities $f_X(\cdot)$ and $f_Y(\cdot)$, respectively. Suppose that $m_X(t)$ and $m_Y(t)$ both exist and are equal for all t . Then the two distribution functions $F_X(\cdot)$ and $F_Y(\cdot)$ are equal.

Exercise 2.6.1: Find the MGF of the binomial distribution

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

2.7 Other Distribution Summaries

Definition 2.15 (Quantile): For a $RV X$, the q^{th} quantile of η_q is the smallest number η satisfying

$$F_X(\eta) \geq q$$

Certain quantiles are often used in statistics and are given particular names: **Median** $\eta_{0.5}$, a measure of the location of the distribution's centre. **Lower Quartile** $\eta_{0.25}$ and **Upper Quartile** $\eta_{0.75}$ can be combined to give a measure of the spread of the distribution termed the **Inter-quartile Range** $\eta_{0.75} - \eta_{0.25}$ and together with the median provide some idea of the skewness of the distribution.

Another measure of the centre of a distribution is its **mode**: the point at which $f_X(\cdot)$ obtains its maximum.

CHAPTER 3

SPECIAL UNIVARIATE DISTRIBUTIONS

3.1. Discrete Distributions

3.1.1 Degenerate Distribution

(Degenerate at k) i.e.

$$P(X = k) = 1 \quad [X \in \{k\}]$$

or $X(\Omega) = k$

Then
$$F(x) = \begin{cases} 0 & \text{if } x < k \\ 1 & \text{if } k \leq x \end{cases}$$

$$E(X) = (k).1 = k, \quad \text{Var}(X) = E(X^2) - E(X)^2$$

$$E(X^2) = k^2.1 = k^2 \quad = k^2 - k^2 = 0$$

$\text{Var}(X) = 0$. This property characterizes degenerate R.V.

3.1.2 Two Point Distribution

$$X \in \{x_1, x_2\} \text{ say } x_1 < x_2$$

Such that

$$P(X = x_1) = p, \quad P(X = x_2) = 1 - p \quad 0 < p < 1$$

Then

$$F(x) = \begin{cases} 0 & \text{if } x < x_1 \\ p & \text{if } x_1 \leq x < x_2 \\ 1 & \text{if } x_2 \leq x \end{cases}$$

$$E(X) = px_1 + (1-p)x_2$$

$$M(t) = pe^{tx_1} + (1-p)e^{tx_2}$$

$$\text{Var}(X) = p(1-p)(x_1 - x_2)^2$$

Special Case:

$$\text{If } x_1 = 1, \quad x_2 = 0$$

Then X is said to be Bernoulli R.V.

$$E(X) = p, \quad \text{Var}(X) = p(1-p)$$

Here, we can call it as probability of success is p and failure is $1-p$

3.1.3 Uniform Distribution on n - Points

X is said to have a uniform distribution on n - points $\{x_1, \dots, x_n\}$ if

$$P(X = x_i) = \frac{1}{n} \quad \forall i = 1, 2, \dots, n$$

Then

$$E(X) = \frac{\sum x_i}{n}$$

$$M(t) = \frac{\sum e^{tx_i}}{n}$$

Special Case

If $x_i = i$

i.e. $X \in \{1, 2, \dots, n\}$

$$\text{Then } E(X) = \frac{n+1}{2}, \quad \text{Var}(X) = \frac{n^2-1}{12}$$

3.1.4 Binomial Distribution

X is said to have binomial with parameter p if its PMF is

$$P(X=k) = {}^nC_k p^k (1-p)^{n-k}; k=0,1,\dots,n$$

$$0 \leq p \leq 1$$

We write

$$X \sim b(n, p)$$

$$E(X) = np$$

$$V(X) = npq$$

$$M(t) = (q + pe^t)^n \quad \forall t$$

Note: Binomial distribution can also be considered as the distribution of sum of n independent & identical Binomial r.v. i.e. $b(1, p)$

Results:

- (i) Let $X_i (i=1,2,\dots,k)$ be independent R.V.'s with $X_i \sim b(n_i, p)$ then $X_1 + X_2 + \dots + X_k \sim b(n_1 + \dots + n_k, p)$ distribution.

In particular, take $n_i = n$ or 1.

- (ii) Let X, Y be two independent, non-negative finite integer valued R.V.'s & let $Z = X + Y$. Then

Z is a binomial R.V. with parameter p iff X & Y are binomial R.V. with same parameter p

- (iii) If $X \sim b(n, p)$ then $n - X \sim b(n, q)$

3.1.5 Negative Binomial Distribution (Pascal or Waiting time distribution)

Let X denotes no. of failures that precede the r^{th} success

Then $X+r$ is total no. of experiments replication to get r successes.

$$\therefore P(X+x) = \binom{x+r-1}{x} p^r (1-p)^x$$

$$x=0,1,2,\dots$$

We write,

$$X \sim NB(r, p)$$

$$E(X) = \frac{rq}{p}; \quad M(t) = p^r (1-qe^t)^{-r}$$

$$Var(X) = \frac{rq}{p^2} \text{ provided } qe^t < 1$$

Note: If we are interested in no. of experiment trials needed.

Then $Y = X + r$

$$\therefore P(Y = y) = \binom{y-1}{r-1} p^r (1-p)^{y-r}; y = r, r+1, \dots$$

Special Case:

If $r = 1$

then we say X is a geometric R.V.

Results:

(i) Let $X_1, \dots, X_k$ be independent $NB(r_i; p)$

then $\sum_{i=1}^k X_i$ follows $NB(r_1 + \dots + r_k, p)$

Corollary: Take $r_i = 1$ then sum of independent geometric is binomial.

(ii) Let X and Y be independent $NB(r_1; p)$ and $NB(r_2; p)$. Then condition PMF of X given $X+Y=t$ is

$$P(X=x|X+Y=t) = \frac{\binom{x+r_1-1}{x} \binom{t+r_2-x-1}{t-x}}{\binom{t+r_1+r_2-1}{t}}$$

In particular, if $r_1 = r_2 = 1$, then this distribution is uniform on t points.

(iii) If X has a geometric distribution then for any $m, n \in \mathbb{N} \cup \{0\}$

$$(\text{Memory Less}) P\{X > m+n | X > m\} = P\{X \geq n\}$$

Converse is also true.

(iv) Let X be a non-negative integer valued R.V. satisfying

$$P\{X > m+1 | X > m\} = P\{X \geq 1\} \quad \forall m$$

Then X must have geometric distribution

(v) Let $X_i (i=1, \dots, n)$ be independent geometric R.V. with parameter p_i

then $X_{(1)} = \min_i \{X_i\}$ is also geometric with parameter $p = 1 - \prod_{i=1}^n (1-p_i)$

3.1.6 Hyper-Geometric Distribution

A box contains N marbles of which M are marked. Now, n marbles are drawn.

Let X denote no. of marked marbles drawn.

$$P(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}; x \leq \min(M, n)$$

$$\max(0, M+n-N) \leq x$$

$$E(X) = \frac{n}{N} M$$

$$Var(X) = \frac{nM}{N^2} (N-M)(N-n)$$

Result:

- (i) Let X and Y be independent R.V.'s with distributions $b(m, p)$ & $b(n, p)$. Then conditional distribution of X given $X + Y$ is hyper geometric.

$$\begin{aligned} P(X=x|X+Y=N) &= \frac{P(X=x, X+Y=N)}{P(X+Y=N)} \\ &= \frac{P(X=x)P(Y=N-x)}{P(X+Y=N)} \\ &= \frac{{}^m C_x p^x q^{m-x} {}^n C_{N-x} p^{N-x} q^{n-N+x}}{{}^{m+n} C_N p^N q^{m+n-N}} \\ &= \frac{{}^m C_x {}^n C_{N-x}}{{}^{m+n} C_N} \end{aligned}$$

3.1.7 Poisson Distribution

A R.V. is said to be a Poisson R.V. with parameter $\lambda > 0$ if its PMF is given by

$$P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}, k=0,1,2,\dots$$

$$E(X) = \lambda, \quad M(t) = e^{\lambda(e^t - 1)}$$

$$\text{Var}(X) = \lambda$$

Results:

- (i) Let $X_1, X_2, \dots, X_n$ be independent Poisson R.V.'s
 $X_k \sim P(\lambda_k), k=1,2,\dots,n$

Then

$$S_n = X_1 + \dots + X_n \text{ is a } P(\lambda_1 + \dots + \lambda_n) \text{ R.V.'s}$$

Converse is also true.

- (ii) Let X & Y be independent R.V.'s with $P(\lambda_1)$ and $P(\lambda_2)$ respectively then conditional distribution of X given $X+Y$ is binomial. (Converse is true)

Uses of the Poisson Distribution

For large n , and small p , $X \sim \text{Bin}(n, p)$ is approximately distributed as $\text{Poi}(np)$. This is sometimes termed the "law of small numbers".

A Poisson Process with rate λ per unit time is such that

- (i) X , the number of occurrences of an event in any given time interval of length t is $\text{Poi}(\lambda t)$.
- (ii) The number of events in non-overlapping time intervals are independent random variables (see later).

3.1.8 Multi-Nomial Distribution (Generalized Binomial Distribution)

Let $x_1, \dots, x_{k-1}$ be non-negative integers such that $x_1 + \dots + x_{k-1} \leq n$ then probability that exactly x_i trials terminate in A_i ($i=1, \dots, k-1$) & hence that $x_k = n - (x_1 + \dots + x_{k-1})$ trials is

$$\frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

$$\sum_{i=1}^k x_i = n$$

$$M(t) = (p_1 e^{t_1} + p_2 e^{t_2} + \dots + p_{k-1} e^{t_{k-1}} + p_k)^n$$

$$E(e^{t_1 X_1 + \dots + t_{k-1} X_{k-1}}) \quad \forall t_1, \dots, t_{k-1} \in \mathbb{R}$$

$$E(X_j) = np_j, \quad \text{Var}(X_j) = np_j(1-p_j)$$

$$\text{Cov}(X_j, X_k) = -np_j p_k$$

Summary

Sl.	Distribution	PMF	$E(X)$	$\text{Var}(X)$	$M(t)$
1	Poisson $X \sim P(\lambda)$	$P(X=k) = e^{-\lambda} \frac{\lambda^k}{k!}$ $k=0,1,2,\dots$	λ	λ	$e^{\lambda(e^t-1)}$
2	Binomial $X \sim B(n, p)$	$P(X=k) = {}^n C_k p^k q^{n-k}$ $k=0,1,2,\dots,n$	np	npq	$(q + pe^t)^n$
3	Uniform	$P(X=k) = \frac{1}{n}$ $k=1,2,\dots,n$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$	$\frac{\sum_{k=1}^n e^{kt}}{n}$
4	Two Point	$P(X=a) = p$ $P(X=b) = 1-p$	$ap+b(1-p)$	$p(1-p)(a-b)^2$	$pe^{at} + (1-p)$
5	Negative Binomial $X \sim NB(r, p)$	$P(X=x) = \binom{x+r-1}{x} p^r (1-p)^x$	$\frac{rq}{p}$	$\frac{rq}{p^2}$	$p^r (1-qe^t)^{-r}$; $qe^t < 1$
6	Geometric $X \sim G(1, p)$	$P(X=x) = pq^x$	$\frac{q}{p}$	$\frac{q}{p^2}$	$\frac{p}{1-qe^t}; qe^t < 1$
7	Hyper-geometric	$P(X=x) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$	$\frac{n}{N} M$	$\frac{nM}{N^2(N-1)} \left[\frac{(N-M)}{(N-n)} \right]$	

Put Your Own Notes

3.2. Continuous Distributions

3.2.1. Uniform Distribution

X is said to have uniform distribution on $[a, b]$ if its PDF is given by

$$f(x) = \begin{cases} \frac{1}{b-a}; & a \leq x \leq b \\ 0; & \text{otherwise} \end{cases}$$

$$E(X) = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

$$M(t) = \frac{1}{t(b-a)}(e^{tb} - e^{ta}); t \neq 0$$

Results: Let X be an R.V. with a continuous D.F. F , then $F(X)$ has the uniform distribution on $[0, 1]$

3.2.2. Gamma Distribution

An R.V. X is said to have gamma distribution with parameters α and β if its PDF is

$$f(x) = \begin{cases} \frac{1}{(\Gamma\alpha)(\beta^\alpha)} x^{\alpha-1} e^{-x/\beta}; & 0 < x \\ 0; & \text{otherwise} \end{cases}$$

We write $X \sim G(\alpha, \beta)$

$$E(X) = \alpha\beta \quad \text{Var}(X) = \alpha\beta^2$$

$$M(t) = \frac{1}{(1-\beta t)^\alpha}, \quad t < \frac{1}{\beta}$$

Special Case:

(a) When $\alpha = 1$, then we say X follows exponential distribution with parameter β

$$f(x) = \begin{cases} \beta^{-1} e^{-x/\beta}; & x > 0 \\ 0; & x < 0 \end{cases}$$

$$\text{or } \lambda = \frac{1}{\beta}$$

$$f(x) = \begin{cases} \lambda e^{-\lambda x}; & x > 0 \\ 0; & \text{otherwise} \end{cases}$$

$$E(X) = \beta = \frac{1}{\lambda}$$

(b) When $\alpha = \frac{n}{2}$ ($n > 0$ integer) & $\beta = 2$

Then

$$f(x) = \begin{cases} \frac{1}{\Gamma_{1/2} 2^{n/2}} e^{-x/2} x^{n/2-1}; & 0 < x < \infty \end{cases}$$

Put Your Own Notes

is said chi-square $\chi^2(n)$ distribution.

$$E(X) = n, \quad Var(X) = 2n$$

$$M(t) = \frac{1}{(1-2t)^{-n/2}}, \quad t < \frac{1}{2}$$

Results: Let $X_i (i=1, \dots, n)$ be independent R.V. such that $X_i \sim G(\alpha_i, \beta)$, then

$$S_n = \sum_{i=1}^n X_i \sim G(\sum \alpha_i, \beta) \text{ R.V.}$$

Corollary:

(i) Take $\alpha_i = 1 \quad \forall i$

Then $S_n \sim G(n, \beta)$

i.e. sum of exponential is Gamma

(ii) Let $X \sim G(\alpha_1, \beta)$ & $Y \sim G(\alpha_2, \beta)$ be independent R.V. then

$X + Y$ and $\frac{X}{Y}$ are independent

or $X + Y$ and $\frac{X}{X+Y}$ all independent

conversely also true.

(iii) Memory less property of exponential

$$P(X > r+s | X > s) = P(X > r)$$

where $X \sim \exp(\lambda)$

(iv) If X & Y are independent exponential R.V.'s with parameter β , then

$$Z = \frac{X}{X+Y} \text{ has a } U(0,1) \text{ distribution}$$

3.2.3. Beta Distribution

An R.V. X is said to have beta distribution with parameters α & β ($\alpha > 0, \beta > 0$) if its PDF is

$$f(x) = \begin{cases} \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

We write $X \sim B(\alpha, \beta)$

$$E(X) = \frac{\alpha}{\alpha + \beta}$$

$$V(X) = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

Note: If $\alpha = \beta = 1$, we have $U(0,1)$

Results:

(i) $X \sim B(\alpha, \beta)$ then $1 - X \sim B(\beta, \alpha)$

(ii) Let $X \sim G(\alpha_1, \beta)$ & $Y \sim G(\alpha_2, \beta)$ be independent then

$$\frac{X}{X+Y} \sim B(\alpha_1, \alpha_2) \text{ R.V.}$$

3.2.4. Normal Distribution (Gaussian Law)

- (a) An R.V. X is said to have a standard normal distribution if its PDF is

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, -\infty < x < \infty$$

we write $X \sim N(0,1)$

- (b) An R.V. is said to have normal distribution with parameters μ & $\sigma(>0)$ if

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < \infty$$

We write $X \sim N(\mu, \sigma^2)$

$$M(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

Central moments

i.e. $E((X-\mu)^n) = 0$ if n is odd

$$= [(2n-1)(2n-3)\dots 3.1] \sigma^{2n} \text{ if } n \text{ is even}$$

Results:

- (i) Let $X_1, \dots, X_n$ be independent R.V.'s such that

$$X_k \sim N(\mu_k, \sigma_k^2), \quad k = 1, \dots,$$

then

$$S_n = \sum_{k=1}^n X_k \text{ is } N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

Corollary:

- (a.) $X_i \sim N(\mu, \sigma^2)$

Then

$$S_n \sim N(n\mu, n\sigma^2)$$

$$\& \frac{S_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- (b.) If $X_i \sim N(0,1) (i=1, \dots, n)$ are independent, then

$$\frac{S_n}{\sqrt{n}} \sim N(0,1)$$

- (ii) Let X & Y be independent R.V.'s then $X+Y$ is normal iff X & Y is normal
- (iii) Let X & Y be independent R.V.'s with $N(0,1)$ then $X+Y$ and $X-Y$ are independent.

Put Your Own Notes

(iv) Let X_1 & X_2 are independent $N(\mu_1, \sigma^2)$ & $N(\mu_2, \sigma^2)$ then $X_1 - X_2$ and $X_1 + X_2$ are independent.

(v) (a) $X \sim N(0,1) \Rightarrow X^2 \sim \chi^2(1)$

(b) $X \sim N(\mu, \sigma^2) \Rightarrow aX \sim N(a\mu, a^2\sigma^2)$

$$aX + b \sim N(a\mu + b, a^2\sigma^2)$$

(vi) $X \sim N(\mu, \sigma^2) \Rightarrow \frac{X - \mu}{\sigma} \sim N(0,1)$

(vii) X & Y be i.i.d. $N(0, \sigma^2)$ R.V.'s then

$$\frac{X}{Y} \sim \text{Cauchy}(1, 0)$$

$$\frac{X}{|Y|} \sim \text{Cauchy}(1, 0)$$

$$\frac{|X|}{|Y|} \text{ has PDF } \frac{2}{\pi(1+z^2)}, 0 < z < \infty.$$

3.2.5. Cauchy Distribution

An R.V. is said to have Cauchy distribution with parameters μ and θ if its PDF is

$$f(x) = \frac{\mu}{\pi} \cdot \frac{1}{\mu^2 + (x - \theta)^2}; -\infty < x < \infty$$

$$\mu > 0$$

We write $X \sim C(\mu, \theta)$

Results:

(i) Moments of order ≥ 1 doesn't exist.

(ii) $X \sim C(\mu_1, \theta_1)$ & $Y \sim C(\mu_2, \theta_2)$ be independent then
 $X + Y \sim \ell(\mu_1 + \mu_2, \theta_1 + \theta_2)$

(iii) $X \sim C(1, 0)$ iff $\frac{1}{X} \sim C(1, 0)$

(iv) If $X \sim U\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ R.V. then $Y = \tan X$ is Cauchy R.V.

CHAPTER 4

JOINT AND CONDITIONAL DISTRIBUTIONS

4.1. Joint Distributions

Definition 4.1 (Joint Distribution Function): For random variables $X_1, \dots, X_k$ all defined on the same $(\Omega, \mathcal{A}, \mathbf{P}[\cdot])$, the function $F: \mathbb{R}^k \rightarrow [0, 1]$

$$F_{X_1, \dots, X_k}(x_1, \dots, x_k) = \mathbf{P}[X_1 \leq x_1; \dots; X_k \leq x_k] \quad \forall (x_1, \dots, x_k)$$

is called the joint distribution function.

Definition 4.2 (Marginal Distribution Function): For $F_{X_1, \dots, X_k}$ and $X_{i1}, \dots, X_{in}$ a strict subset of $X_1, \dots, X_k$, the function $F_{X_{i1}, \dots, X_{in}}$ is called a marginal distribution function.

Definition 4.3 (Joint Discrete Density Function): For a k -dimensional discrete random variable $(X_1, \dots, X_k)$, the function

$$f_{X_1, \dots, X_k}(x_1, \dots, x_k) = \mathbf{P}[X_1 = x_1; \dots; X_k = x_k] \quad \forall (x_1, \dots, x_k)$$

is called the joint discrete density function, joint density function, or joint probability function.

Definition 4.4 (Marginal Discrete Density Function): For the joint discrete density function $f_{X_1, \dots, X_k}$ and $X_{i1}, \dots, X_{in}$ a strict subset of $X_1, \dots, X_k$, the function $f_{X_{i1}, \dots, X_{in}}$ is called a marginal discrete density function.

Definition 4.5 (Joint Continuous Density Function): For a k -dimensional random variable $(X_1, \dots, X_k)$, the function $f_{X_1, \dots, X_k}(x_1, \dots, x_k) \geq 0$ such that

$$f_{X_1, \dots, X_k}(x_1, \dots, x_k) = \int_{-\infty}^{x_k} \dots \int_{-\infty}^{x_1} f_{X_1, \dots, X_k}(u_1, \dots, u_k) du_1 \dots du_k$$

for all $(x_1, \dots, x_k)$ is called a joint probability density function.

Definition 4.6 (Marginal Density Function): For the joint continuous density function $f_{X_1, \dots, X_k}$ and $X_{i1}, \dots, X_{in}$ a strict subset of $X_1, \dots, X_k$, the function $f_{X_{i1}, \dots, X_{in}}$ is called a marginal density function.

4.2. Special Multivariate Distributions

There are many ways to construct multivariate distributions and a great many named multivariate distributions exist. This section provides a brief

summary of a small number of the most ubiquitous multivariate distributions.

4.2.1. Multinomial Distribution

The multivariate distribution is a generalisation of the binomial distribution.

$$f_{X_1, \dots, X_k}(x_1, \dots, x_k) = \text{Mult}(x; n, p_1, \dots, p_{k+1}) = \frac{n!}{\prod_{i=1}^{k+1} x_i!} \prod_{i=1}^{k+1} p_i^{x_i}$$

Where $x_i = 0, \dots, n$ and $\sum_{i=1}^{k+1} x_i = n$. (n is fixed, so the value of x_{k+1} is determined by values of $x_1, \dots, x_k$).

Note that the i^{th} element of $\mathbf{X}$, X_i , has a $\text{Bin}(n, p_i)$ marginal distribution.

4.2.2. Bi-Variate Normal Distribution

The normal distribution can be straightforwardly extended to a bivariate random variable by considering

to random variables which are correlated and which each is marginally normal:

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 - 2\rho \left(\frac{x_1 - \mu_1}{\sigma_1} \right) \left(\frac{x_2 - \mu_2}{\sigma_2} \right) \right] \right\}$$

for $-\infty < x_1, x_2, \mu_1, \mu_2 < \infty, \sigma_1, \sigma_2 > 0, -1 < \rho < 1$.

- (a) ρ is the correlation coefficient
- (b) for $\rho = 0$ the bi-variate normal density is the product of two uni-variate normal densities: this corresponds to X_1 and X_2 being independent Normal random variables.
- (c) $x + y$ and $x - y$ are independent iff $\sigma_1^2 = \sigma_2^2$
- (d) x and y are independent iff $\rho = 0$

Extension to Multivariate Normal Distribution: In fact, it's straightforward to extend the normal distribution to vectors or arbitrary length, the multivariate normal distribution has density.

$$f(x; \mu, \Sigma) = N(x; \mu, \Sigma) = \left(\frac{1}{\sqrt{2\pi}} \right)^r |\Sigma|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right\}$$

Where

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_r \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_r \end{pmatrix}, \quad \Sigma = \mathbf{E} \left[(X - \mu)(X - \mu)^T \right]$$

Note that x is a vector; it has mean μ which is itself a vector and Σ is the variance-covariance matrix. If x is k -dimensional then Σ is a $k \times k$ matrix.

4.3. Conditional Distributions and Densities

Given several random variables how much information does knowing one provide about the others? The notion of conditional probability provides an explicit answer to this question.

Definition 4.7 (Conditional Discrete Density Function): For discrete random variables with X and Y with probability mass points $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$,

$$f_{Y|X}(y_j | x_i) = \frac{P[X = x_i; Y = y_j]}{P[X = x_i]} = P[Y = y_j | X = x_i]$$

is called the conditional discrete density function of Y given $X = x$.

Definition 4.8 (Conditional Discrete Distribution): For jointly discrete random variables X and Y ,

$$f_{Y|X}(y | x) = P[Y \leq y | X = x] = \sum_{j: y_j \leq y} f_{Y|X}(y_j | x)$$

is called the conditional discrete distribution of Y given $X = x$.

Definition 4.9 (Conditional Probability Density Function): For continuous random variables X and Y with joint probability density function $f_{X,Y}(x, y)$,

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}, \text{ if } f_X(x) > 0$$

Where $f_X(x)$ is the marginal density of X .

Conditional Distribution For jointly continuous random variables X and Y ,

$$f_{Y|X}(y | x) = \int_{-\infty}^y f_{Y|X}(z | x) dz \quad \forall x \text{ such that } f_X(x) > 0.$$

4.4. Conditional Expectation

We can also ask what the expected behaviour of one random variable is, given knowledge of the value of a second random variable and this gives rise to the idea of conditional expectation.

Definition 4.10 (Conditional Expectation): The conditional expectation in discrete and continuous cases corresponds to an expectation with respect to the appropriate conditional probability distribution:

Discrete

$$E[Y | X = x] = \sum_{\text{all } y} y P_{Y|X}(Y = y | X = x)$$

Continuous

$$E[Y | X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(Y | X = x) dy.$$

Note that before X is known to take the value x , $E[Y | X]$ is itself a random variable being a function of the random variable X . We might be interested in the distribution of the random variable $E[Y | X]$.

Theorem 4.1 (Tower Property of Conditional Expectation): For any two random variables X_1 and X_2

$$E[E[X_1 | X_2]] = E[X_1]$$

Exercise 4.4.1: Suppose that $\Theta \sim U[0, 1]$ and $(X | \Theta) \sim \text{Bin}(2, \Theta)$.

Find $E[X | \Theta]$ and hence or otherwise show that $E[X] = 1$.

4.5. Conditional Expectations of Functions of Random Variables

By extending the theorem on marginal expectations we can relate the conditional and marginal expectations of functions of random variables (in particular, their variances).

Theorem 4.2: (Marginal Expectation of a Transformed Random Variables): For any random variables X_1 and X_2 , and for any function $h(\cdot)$,

$$E[E[h(X_1) | X_2]] = E[h(X_1)].$$

Theorem 4.3 (Marginal Variance): For any random variables X_1 and X_2 ,

$$\text{Var}(X_1) = E[\text{Var}(X_1 | X_2)] + \text{Var}(E[X_1 | X_2]).$$

4.6. Independence of Random Variables

Whilst the previous sections have been concerned with the information that one random variable carries about another, it would seem that there must be pairs of random variables which each provide no information whatsoever about the other. It is, for example, difficult to imagine that the value obtained when a die is rolled in Coventry will tell us much about the outcome of a coin toss taking place at the same time in Lancaster.

There are two equivalent statements of a property termed stochastic independence which capture precisely this idea. The following two definitions are equivalent for both discrete and continuous random variables.

Definition 4.11: (Stochastic Independence): Definition 1 Random variables $X_1, X_2, \dots, X_n$ are stochastically independent iff

$$F_{X_1, \dots, X_n}(x_1, \dots, x_k) = \prod_{i=1}^n F_{X_i}(x_i)$$

Definition 2: Random variables $X_1, X_2, \dots, X_n$ are stochastically independent iff

$$f_{X_1, \dots, X_n}(x_1, \dots, x_k) = \prod_{i=1}^n f_{X_i}(x_i)$$

If X_1 and X_2 are independent then their conditional densities are equal to their marginal densities

4.7. Covariance and Correlation

Having established that sometimes one random variable does convey information about another and in other cases knowing the value of a random variable tells us nothing useful about another random variable it is useful to have mechanisms for characterising the relationship between pairs (or larger groups) of random variables.

Definition 4.12 (Covariance and Correlation): Covariance: For random variables X and Y defined on the same probability space

$$\text{Cov}[X, Y] = E[(X - \mu_X)(Y - \mu_Y)]$$

$$= E[XY] - \mu_X \mu_Y$$

Correlation: For random variables X and Y defined on the same probability space

$$\rho[X, Y] = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y} = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X] \text{Var}[Y]}}$$

provided that $\sigma_X > 0$ and $\sigma_Y > 0$.

Theorem 4.4 (Cauchy-Schwarz Inequality): Let X and Y have finite second moments. Then

$$(E[XY])^2 = |E[XY]|^2 \leq E[X^2] E[Y^2]$$

with equality if and only if $P[Y = cX] = 1$ for some constant c .

4.8. Transformation of Random Variables: $Y = g(X)$

Theorem 4.5 (Distribution of a Function of a Random Variable): Let X be a random variable and $Y = g(X)$ where g is injective (i.e. it maps at most one x to any value y). Then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|$$

given that $(g^{-1}(y))'$ exists and $(g^{-1}(y))' > 0 \forall y$ or $(g^{-1}(y))' < 0 \forall y$. If g is not bijective (one-to-one) there may be values of y for which there exists no x such that $y = g(x)$. Such points clearly have density zero.

When the conditions of this theorem are not satisfied it is necessary to be a little more careful. The most general approach for finding the density of a transformed random variable is to explicitly construct the distribution function of the transformed random variable and then to use the standard approach to turn the distribution function into a density (this approach is discussed in Larry Wasserstein's "All of Statistics").

Exercise 4.8.1: Let X be distributed exponentially with parameter α , that is

$$f_X(x) = \begin{cases} \alpha e^{-\alpha x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Find the density function of

$$(i) \quad Y = g(X) \text{ with } g(X) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - e^{-\alpha x} & \text{for } x \geq 0 \end{cases}$$

$$(ii) \quad Y = X^{1/\beta}, \beta > 0$$

$$(iii) \quad Y = g(X) \text{ with } g(X) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x \leq 1 \\ 1 & \text{for } x > 1 \end{cases}$$

Theorem 4.6 (Probability Integral Transformation): If X is a random variable with continuous $F_X(x)$, then $U = F_X(X)$ is uniformly distributed over the interval $(0, 1)$.

Conversely if U is uniform over $(0, 1)$, then $X = F_X^{-1}(U)$ has distribution function F_X .

4.9. Moment-Generating-Function Technique

The following technique is but one example of a situation in which the moment generating function proves invaluable.

Function of a Variable. For $Y = g(X)$ compute

$$m_Y(t) = E[e^{tY}] = E[e^{tg(X)}]$$

If the result is the MGF of a known distribution then it will follow that Y has that distribution.

Sums of Independent random variables. For $Y = \sum_i X_i$, where the X_i are independent random variables for which the MGF exists $\forall -h < t < h, h > 0$

$$m_Y(t) = E[e^{\sum_i tX_i}] = \prod_i m_{X_i}(t) \quad \text{for } -h < t < h$$

Thus $\prod_i m_{X_i}(t)$ may be used to identify the distribution of Y as above.

CHAPTER 5

INFERENCE

5.1. Sample Statistics

Suppose we select a sample of size n from a population of size N . For each i in $\{1, \dots, n\}$, let X_i be a random variable denoting the outcome of the i^{th} observation of a variable of interest. For example, X_i might be the height of the i^{th} person sampled. Under the assumptions of simple random sampling, the X_i are independent and identically distributed (*iid*).

Therefore, if the distribution of a single unit sampled from the population can be characterized by a distribution with density function f , the marginal density function of each X_i is also f and their joint density function g is a simple product of their marginal densities:

$$g(x_1, x_2, \dots, x_n) = f(x_1)f(x_2)\dots f(x_n)$$

In order to make inferences about a population parameter, we use sample data to form an estimate of the population parameter. We calculate our estimate using an estimator or sample statistic, which is a function of the X_i . We have already seen examples of sample statistics, for example the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

where n is the size of the sample is an estimator of the population mean, e.g. for discrete X

$$\mu_X = \sum_{j=1}^N x_j \mathbb{P}[X = x_j]$$

where N is the number of distinct values which it is possible for an X_i to take.

5.2. Sampling Distributions

Since an estimator $\hat{\theta}$ is a function of random variables, it follows that $\hat{\theta}$ is itself a random variable and possesses its own distribution. The probability distribution of an estimator itself is called a sampling distribution.

Proposition 5.1 (Distribution of The Sample Mean): Let $\bar{X}$ denote the sample mean of a random sample of size n from a normal distribution with mean μ and variance σ^2 . Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Theorem 5.1 (Central Limit Theorem): Let f be a density function with mean μ and finite variance σ^2 . Let $\bar{X}$ be the sample mean of a random sample of size n from f and let

$$Z_n = \frac{\bar{X} - E[\bar{X}]}{\sqrt{\text{Var}[\bar{X}]}} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Then the distribution of Z_n approaches the standard normal distribution as $n \rightarrow \infty$. This is often written as: $Z_n \xrightarrow{d} N(0, 1)$ with $\xrightarrow{d}$ denoting convergence in distribution.

Thus, if the sample size is "large enough", the sample mean can be assumed to follow a normal distribution regardless of the population distribution. In practice, this assumption is often taken to be valid for a sample size $n > 30$.

The Chi-Squared Distribution

The chi-squared distribution is a special case of the gamma distribution. The sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

of a standard normal distribution is χ^2 with $n-1$ degrees of freedom.

Definition 5.1: If X is a random variable with density

$$f(x) = \chi_k(x) = \begin{cases} \frac{1}{\Gamma\left(\frac{k}{2}\right)} \left(\frac{1}{2}\right)^{\frac{k}{2}} x^{\frac{k}{2}-1} e^{-\frac{1}{2}x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

then X is defined to have a χ^2 distribution with k degrees of freedom (χ_k^2) where k is a positive integer.

Thus the χ_k^2 density is a gamma density with $r = \frac{k}{2}$, $\lambda = \frac{1}{2}$.

Result: If the RVs $X_i, i=1, \dots, n$ are independently normally distributed with means μ_i and variances σ_i^2 then

$$U = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2$$

has a χ_n^2 distribution.

Theorem 5.2: If $X_1, \dots, X_n$ is a random sample from a normal distribution with mean μ and variance σ^2 then

(i.) $\bar{X}$ and $\sum_{i=1}^n (X_i - \bar{X})^2$ are independent,

(ii.) $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2}$ has a χ_{n-1}^2 distribution.

Corollary 5.1: If $s^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$ is the sample variance of a random sample of size n from a normal distribution with mean μ and variance σ^2 , then

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

The t Distribution

The t distribution is closely related to the normal distribution and is needed for making inferences about the mean of a normal distribution when the variance is also unknown.

Definition 5.2: If $Z \sim N(0, 1)$, $U \sim \chi_k^2$ and Z and U are independent of one other, then

$$X = \frac{Z}{\sqrt{\frac{U}{k}}} \sim t_k$$

where t_k denotes a t distribution with k degrees of freedom.

The density of the t_k distribution is:

$$f(x) = t_k(x) = \frac{\Gamma\left[\frac{k+1}{2}\right]}{\Gamma\left[\frac{k}{2}\right]} \frac{1}{\sqrt{k\pi}} \frac{1}{\left(1 + \frac{x^2}{k}\right)^{\frac{k+1}{2}}}$$

For $-\infty < x < \infty$.

The MGF for the t distribution does not exist, but

$$E[X] = 0$$

as the distribution is symmetric (actually the expectation only exists when $k > 1$ although an extension known as the Cauchy principle value can be defined more generally), and it can be shown that

$$\text{Var}[X] = \frac{k}{k-2}, \text{ for } k > 2.$$

Theorem 5.3: If $X \sim t_k$ then

$$f(x) = \frac{\Gamma\left[\frac{k+1}{2}\right]}{\Gamma\left[\frac{k}{2}\right]} \frac{1}{\sqrt{k\pi}} \frac{1}{\left(1 + \frac{x^2}{k}\right)^{\frac{k+1}{2}}} \rightarrow \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

as $k \rightarrow \infty$. That is, as $k \rightarrow \infty$ the density approaches the density of a standard normal.

The F Distribution

The F Distribution is useful for making inferences about the ratio of two unknown variances.

Put Your Own Notes

Definition 5.3: Suppose U and V are independently distributed with $U \sim \chi_m^2$ and $V \sim \chi_n^2$. Then the random variable

$$X = \frac{U}{m} / \frac{V}{n}$$

is distributed according to an F distribution with m and n degrees of freedom.

The density of X is given by

$$f(x) = F_{m,n}(x) = \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} \left(\frac{m}{n}\right)^{\frac{m}{2}} \frac{x^{\frac{m-2}{2}}}{\left[1 + \frac{m}{n}x\right]^{\frac{m+n}{2}}}, \quad x > 0$$

Returning to the case of two independent random samples from normal populations of common variance, but differing means:

$$X_1, \dots, X_{n_1} \sim N(\mu_1, \sigma^2) \quad Y_1, \dots, Y_{n_2} \sim N(\mu_2, \sigma^2)$$

we have

$$\frac{(n_1 - 1)s_X^2}{\sigma^2} \sim \chi_{n_1-1}^2 \quad \text{and} \quad \frac{(n_2 - 1)s_Y^2}{\sigma^2} \sim \chi_{n_2-1}^2$$

so that

$$\frac{\frac{(n_1 - 1)s_X^2}{\sigma^2} / (n_1 - 1)}{\frac{(n_2 - 1)s_Y^2}{\sigma^2} / (n_2 - 1)} = \frac{s_X^2}{s_Y^2} \sim F_{n_1-1, n_2-1}.$$

5.3. Point Estimation

The sample mean and variance are examples of point estimators, because the estimates they produce are single point values, rather than a range of values. For a given parameter there are an infinite number of possible estimators, hence the question arises: what makes a "good" estimator?

Definition 5.4 (Unbiasedness): Let X be a random variable with pdf $f(x; \theta)$, where $\theta \in \Omega \subset \mathbb{R}^p$ is some unknown parameter, $p \geq 1$. Let $X_1, \dots, X_n$ be a random sample from the distribution of X and let $\hat{\theta}$ denote a statistic. $\hat{\theta}$ is an unbiased estimator of θ if

$$E[\hat{\theta}] = \theta \quad \forall \theta \in \Omega$$

where the expectation is with respect to $f(x; \theta)$.

If $\hat{\theta}$ is not unbiased, we say that $\hat{\theta}$ is a biased estimator of θ , with

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

If $\text{Bias}(\hat{\theta}) \rightarrow 0$ when $n \rightarrow \infty$ then we say that $\hat{\theta}$ is asymptotically unbiased.

Example 5.1: Consider the following estimator of the population variance

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

We find that

$$\begin{aligned} E[\hat{\sigma}^2] &= E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right] \\ &= \frac{1}{n} E\left[\sum_{i=1}^n X_i^2 - n\bar{X}^2\right] \\ &= \frac{1}{n} \left(\sum_{i=1}^n E[X_i^2] - nE[\bar{X}^2] \right) \\ &= \frac{1}{n} \left(n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n} + \mu^2\right) \right) \\ &= \frac{n-1}{n} \sigma^2 \end{aligned}$$

So this estimator is biased with bias equal to $-\frac{\sigma^2}{n}$. As this decays to zero as $n \rightarrow \infty$ it is an asymptotically unbiased estimator. However, we can see that the sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Consistency: In order to define a consistent estimator, we first define convergence in probability.

Definition 5.5 (Convergence in Probability): Let $\{X_n\}$ be a sequence of random variables and let X be a random variable. We say that X_n converges in probability to X if $\forall \varepsilon > 0$

$$\lim_{n \rightarrow \infty} P[|X_n - X| \geq \varepsilon] = 0, \text{ or equivalently } \lim_{n \rightarrow \infty} P[|X_n - X| < \varepsilon] = 1.$$

If so, it is common to write $X_n \xrightarrow{P} X$.

Definition 5.6 (Consistent Estimator): Let $X_1, \dots, X_n$ be a sample from the distribution of X where X is a random variable with distribution function $F(x; \theta)$. Let $\hat{\theta}$ denote a statistic. $\hat{\theta}$ is a consistent estimator of θ if, whatever the value of θ ,

$$\hat{\theta} \xrightarrow{P} \theta.$$

A particular case of consistency is often used to justify using sample averages:

Theorem 5.4 (Weak Law of Large Numbers): Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ with $X_1, \dots, X_n$ iid. Then

$$\bar{X} \xrightarrow{P} \mu,$$

i.e. $\bar{X}$ is a consistent estimator of μ .

Put Your Own Notes

Theorem 5.5: Suppose $X_n \xrightarrow{P} a$ and the real function g is continuous at a . Then $g(X_n) \xrightarrow{P} g(a)$.

Consistency according to the definition above may be hard to prove, but it turns out that a sufficient (though not necessary) condition for consistency is that $\text{Bias}(\hat{\theta}) \rightarrow 0$ and $\text{Var}(\hat{\theta}) \rightarrow 0$ as $n \rightarrow \infty$.

Definition 5.7 (Consistency in Mean-Squared Error): If $\hat{\theta}$ is an estimator of θ , then the mean squared error of $\hat{\theta}$ is defined as

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

and $\hat{\theta}$ is said to be consistent in MSE if $\text{MSE}(\hat{\theta}) \rightarrow 0$ as the size of the sample on which $\hat{\theta}$ is based increases to infinity.

Result: $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$.

5.4. Interval Estimation

This section describes confidence intervals which are intervals constructed such that they contain θ with some level of confidence.

Definition 5.8 (Confidence Interval): Let $X_1, \dots, X_n$ be a random sample from a distribution with pdf $f(x; \theta)$ where θ is an unknown parameter in the parameter space Ω . If L and U are statistics such that

$$P[L \leq \theta \leq U] = 1 - \alpha$$

then the interval (L, U) is a $100(1 - \alpha)\%$ confidence interval for θ .

$1 - \alpha$ is known as the confidence coefficient and α is the level of significance.

There are several ways in which confidence intervals can be constructed. The basic procedure we shall use to construct a $100(1 - \alpha)\%$ confidence interval for a parameter is as follows

- (i) Select a sample statistic to estimate the parameter.
- (ii) Identify the sampling distribution for the statistic.
- (iii) Determine the bounds within which the sample statistic will reside with probability $1 - \alpha$.
- (iv) Invert these bounds to obtain an expression in terms of θ .

Confidence Intervals based on the CLT: Suppose that we are interested in the population mean θ and we wish to use the sample mean $\bar{X}$ as an estimator for θ . Then if the population density has variance σ^2 the CLT states that

$$Z_n = \frac{\bar{X} - \theta}{\frac{\sigma}{\sqrt{n}}} \xrightarrow{d} N(0, 1)$$

Where n is the sample size.

Hence $P(-1.96 < Z < 1.96) = 0.95$ and we can construct a 95% confidence interval as follows

$$0.95 = P\left(-1.96 < \frac{\sqrt{n}(\bar{X} - \theta)}{\sigma} < 1.96\right)$$

$$= P\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \theta < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right)$$

In the above derivation it was assumed that the variance σ^2 was known. If the variance is unknown, but we estimate σ with a consistent estimator s , then it can be shown that

$$\frac{\sqrt{n}(\bar{X} - \theta)}{s} \xrightarrow{d} N(0, 1)$$

and hence

$$\bar{X} \pm 1.96 \frac{s}{\sqrt{n}}$$

would be the endpoints of an approximate 95% CI for θ .

More generally the endpoints of a $100(1-\alpha)\%$ CI for θ are given by

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

Where $z_{\frac{\alpha}{2}}$ is such that

$$P\left(Z > z_{\frac{\alpha}{2}}\right) = 1 = \Phi\left(z_{\frac{\alpha}{2}}\right) = \frac{\alpha}{2}$$

The statistic $\frac{s}{\sqrt{n}}$ is known as the standard error.

Note that the confidence interval satisfies a number of intuitively logical requirements:

- a higher confidence coefficient leads to a longer CI,
- a larger sample size leads to a shorter CI,
- a larger variance leads to a longer CI.

We now consider four cases where the CI is found using the CLT approach.

CI for the Mean: Let $X_1, \dots, X_n$ be a random sample from the distribution of an RV X with unknown mean μ and variance σ^2 . Let $\bar{X}$ and s^2 denote the sample mean and variance. We have shown that s^2 is a consistent estimator for σ^2 and hence

$$\frac{\sqrt{n}(\bar{X} - \theta)}{s} \xrightarrow{d} N(0, 1).$$

Thus the endpoints of an approximate $100(1-\alpha)\%$ CI for μ are

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}.$$

CI for a Proportion: Let $X \sim \text{Ber}(p)$ and let $X_1, \dots, X_n$ be a random sample from the distribution of X . Let $\hat{p} = \bar{X}$ be the sample proportion of successes. By the CLT we have

Put Your Own Notes

$$\hat{p} \xrightarrow{d} N\left(p, \frac{p(1-p)}{n}\right).$$

By the weak law of large numbers $\hat{p}$ is a consistent estimator of p and by Theorem 5.6 $\hat{p}(1-\hat{p})$ is a consistent estimator of $p(1-p)$. Hence, an approximate $100(1-\alpha)\%$ CI for p is

$$\hat{p} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Exercise 5.4.1: In a random sample 136 of 400 people given a flu vaccine experienced some discomfort.

Construct a 95% CI for the true proportion of people who will experience discomfort from the vaccine.

CI for Differences in Means: Suppose we have two random variables X and Y and we are interested in comparing their means, μ_1 and μ_2 respectively. We shall derive a CI for $\Delta = \mu_1 - \mu_2$. Let $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$ be random samples from the distributions of X and Y respectively and let $\sigma_1 = \text{Var}[X]$ and $\sigma_2 = \text{Var}[Y]$. Suppose we estimate Δ by the unbiased estimator $\bar{X} - \bar{Y}$ and estimate σ_1 and σ_2 by the appropriate sample variances. Then because

$$\bar{X} \xrightarrow{d} N\left(\mu_1, \frac{s_1^2}{n}\right)$$

and

$$\bar{Y} \xrightarrow{d} N\left(\mu_2, \frac{s_2^2}{n}\right)$$

the endpoints of an approximate $100(1-\alpha)\%$ CI for Δ are

$$\bar{X} - \bar{Y} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

CI for Differences in Proportions: For differences in proportions where $X \sim \text{Ber}(p_1)$ and $Y \sim \text{Ber}(p_2)$ the approximate CI from the CLT is

$$\hat{p}_1 - \hat{p}_2 \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{n}}.$$

Exercise 5.4.2: If 132 of 200 male voters and 90 out of 159 female voters favour a certain candidate running for governor of Illinois, find a 99% CI for the difference between the actual proportions of male and female voters who favour the candidate.

Note that whenever a CI is obtained via the CLT it holds exactly only asymptotically; such confidence intervals are only approximations for finite samples although the approximations can be extremely good for even moderately large samples.

CIs for Normal Populations: We now consider the special case of sampling from normal populations. In this case we can use the exact sampling distribution to derive confidence intervals as shown in the following examples.

CI for the Mean of a Normal Distribution: Suppose $X_1, \dots, X_n$ are a random sample from a $N(\mu, \sigma^2)$ distribution. In Section 5.2 we saw that

$$T = \frac{\sqrt{n}(\bar{X} - \mu)}{s} \sim t_{n-1}.$$

Hence the exact $100(1-\alpha)\%$ CI for μ has endpoints

$$\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}.$$

Remember that $t_{\frac{\alpha}{2}, n-1} \rightarrow z_{\frac{\alpha}{2}}$ as $n \rightarrow \infty$. In practice, if $n \geq 120$, the quantiles

of the standard normal can be used instead (they are essentially indistinguishable from those of the t distribution in this regime).

Exercise 5.4.3: Suppose we observe the following data on lactic acid concentrations in cheese

0.86 1.53 1.57 1.81 0.99 1.09 1.29 1.78 1.29 1.58

Assuming that these data are a random sample from a normal distribution, calculate a 90% CI for the mean of this distribution.

CI for Differences in Means of Normal Populations with Equal Variance: Suppose we have two independent random samples

$$X_1, \dots, X_{n_1} \sim N(\mu_1, \sigma^2) \quad Y_1, \dots, Y_{n_2} \sim N(\mu_2, \sigma^2).$$

In Section 5.2 we saw that

$$T = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2},$$

Where s_p^2 is the pooled variance

$$s_p^2 = \frac{(n_1 - 1)s_X^2 + (n_2 - 1)s_Y^2}{n_1 + n_2 - 2}$$

Which can be shown to be an unbiased estimator of σ^2 .

Therefore the exact $100(1-\alpha)\%$ CI for $\mu_1 - \mu_2$ in this case is

$$\bar{X} - \bar{Y} \pm t_{\frac{\alpha}{2}, n_1+n_2-2} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} s_p.$$

In the case of unequal variances, an exact CI cannot be derived.

Exercise 5.4.4: Consider a study to compare the nicotine contents of two brands of cigarettes. 10 cigarettes of Brand A had an average nicotine content of 3.1 milligrams with a standard error of 0.5 mg, while 8 cigarettes of Brand B had an average nicotine content of 2.7 mg with a standard error of 0.7 mg. Assuming that the 2 data sets are independent random samples from normal populations with equal variances, construct a 95% confidence interval for the difference between the mean nicotine contents of the two brands.

Confidence Interval for the Variance of a Normal Population: Suppose $X_1, \dots, X_n$ are i.i.d normal with unknown μ and σ^2 . From the Corollary of Theorem 5.3 we know that

$$Q = \frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Put Your Own Notes

Therefore

$$P\left(\chi_{1-\alpha/2, n-1}^2 < \frac{(n-1)s^2}{\sigma^2} < \chi_{\alpha/2, n-1}^2\right) = 1 - \alpha$$

Where $P\left(Q < \chi_{\alpha/2, n-1}^2\right) = 1 - \frac{\alpha}{2}$. Hence an exact $100(1-\alpha)\%$ CI for σ^2 is given by

$$\left(\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}\right)$$

Note that the χ^2 distribution is asymmetric, hence this CI is asymmetric, unlike the previous CIs we have defined. This illustrates an important point: any interval which has the appropriate coverage probability can be used as a confidence interval and there is no unique $(1-\alpha)$ confidence interval for a statistic (imagine shifting the left endpoint of the interval slightly towards the centre; the right endpoint could then be moved a small amount away from the centre to compensate). Symmetry is one of the properties which is often considered desirable when dealing with confidence intervals, but it's certainly not essential.

CI for the Ratio of Variance of Normal Populations: Suppose we have two independent random samples

$$X_1, \dots, X_{n_1} \sim N(\mu_1, \sigma_1^2) \quad Y_1, \dots, Y_{n_2} \sim N(\mu_2, \sigma_2^2)$$

Where μ_1, μ_2, σ_1 and σ_2 are unknown. Let

$$Q = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2}$$

In Section 5.2 we found that

$$\frac{s_X^2}{s_Y^2} \sim F_{n_1-1, n_2-1}$$

Thus we have

$$P\left(F_{n_1-1, n_2-1, 1-\alpha/2} < Q < F_{n_1-1, n_2-1, \alpha/2}\right) = 1 - \alpha$$

$$P\left(\frac{1}{F_{n_2-1, n_1-1, \alpha/2}} < Q < F_{n_1-1, n_2-1, \alpha/2}\right) = 1 - \alpha$$

from which we derive the following exact CI for $\frac{\sigma_1^2}{\sigma_2^2}$

$$\left(\frac{s_1^2}{s_2^2} \frac{1}{F_{n_2-1, n_1-1, \alpha/2}}, \frac{s_1^2}{s_2^2} F_{n_1-1, n_2-1, \alpha/2}\right)$$

Note this CI is also asymmetric.

Example 5.2: Consider the study described in Exercise 5.4.4. A 95% CI for the ratio of the variances

of the two populations is given by

$$\left(\frac{0.5^2}{0.7^2} \frac{1}{F_{7,9,0.025}}, \frac{0.5^2}{0.7^2} F_{9,7,0.025}\right) = \left(\frac{0.5^2}{0.7^2} \frac{1}{4.197}, \frac{0.5^2}{0.7^2} (4.823)\right) = (0.122, 2.46)$$

Which includes 1, so the assumption of equal variances seems reasonable in this case.

Put Your Own Notes

CHAPTER 6

MAXIMUM LIKELIHOOD ESTIMATION

6.1. Likelihood Function and ML Estimator

Suppose we have a simple random sample $X_1, \dots, X_n$ from a density $f(x; \theta)$ parameterized by some possibly unknown parameter θ .

The joint pdf of the entire data set is

$$f(x; \theta) = \prod_{i=1}^n f(x_i; \theta)$$

with $x = (x_1, \dots, x_n)^T$.

The likelihood function $L(\theta; x)$ is this joint pdf viewed as a function of θ with x fixed to the observed data. It is often more convenient to work with the log-likelihood

$$l(\theta; x) = \log[L(\theta; x)]$$

Which in the simple random sampling case above yields:

$$l(\theta; x) = \sum_{i=1}^n \log f(x_i; \theta).$$

The maximum likelihood estimator (MLE) is the value $\hat{\theta}$ which maximises $L(\theta; x)$. The MLE also maximises $l(\theta; x)$ because $\log(\cdot)$ is a monotonic function. Usually it is easier to maximize $l(\theta; x)$, so we work with this.

Example 6.1: Suppose $X_1, \dots, X_n$ is a random sample from the exponential distribution with pdf

$$f(x; \theta) = \begin{cases} \theta e^{-\theta x} & x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

$$L(\theta; x) = \theta^n e^{-\theta \sum_{i=1}^n x_i}, \text{ so } l(\theta; x) = n \log(\theta) - \theta \sum_{i=1}^n x_i$$

$$\frac{\partial l}{\partial \theta} = \frac{n}{\theta} - \sum_{i=1}^n x_i = 0 \text{ gives the MLE } \hat{\theta} = \frac{1}{\bar{X}}$$

Check: $\frac{\partial^2 l}{\partial \theta^2} = -\frac{n}{\theta^2} < 0$, so $\hat{\theta}$ does correspond to a maximum.

Exercise 6.2: Suppose $X_1, \dots, X_n$ form a random sample from the Poisson distribution $\text{Poisson}(\theta)$. Derive the ML estimator for θ .

6.4. Properties of MLEs

The maximum likelihood estimator has a number of useful properties.

Theorem 6.1: Suppose θ and ϕ represent two alternative parameterizations and that ϕ is a one-to-one function of θ (more formally, there exists a bijective mapping, g , from θ to ϕ), so we can write

$$\phi = g(\theta), \theta = h(\phi)$$

for appropriate g and $h = g^{-1}$.

Then if $\hat{\theta}$ is the MLE of θ , then the MLE of ϕ is $g(\hat{\theta})$.

Corollary 6.1 (Invariance of MLE): Let $\hat{\theta}_1, \dots, \hat{\theta}_k$ be a MLE for $\theta_1, \dots, \theta_k$. If

$$T(\theta) = (T_1(\theta), \dots, T_r(\theta))$$

is a sufficiently regular transformation of the PARAMETER space Ω , then

$$T(\hat{\theta}) = (T_1(\hat{\theta}), \dots, T_r(\hat{\theta}))$$

is a MLE of $T(\theta)$.

Lemma 6.1: Suppose there exists an unbiased estimator, $\tilde{\theta}$, which attains the Cramer-Rao bound. Suppose that $\hat{\theta}$, the MLE is a solution to

$$\frac{\partial l}{\partial \theta} = 0.$$

Then $\tilde{\theta} = \hat{\theta}$.

Put Your Own Notes

CHAPTER 7

HYPOTHESIS TESTING

7.1. Introduction

A hypothesis is a falsifiable claim about the real world; a statement which if true will explain some observable quantity. Statisticians will be interested in hypotheses like:

- (i) "The probabilities of a male panda or a female panda being born are equal",
- (ii) "The number of flying bombs falling on a given area of London during World War II follows a Poisson distribution",
- (iii) "The mean systolic blood pressure of 35-year-old men is no higher than that of 40-year-old women",
- (iv) "The mean value of $Y = \log(\text{systolic blood pressure})$ is independent of $X = \text{age}$ " (i.e. $E[Y | X = x] = \text{constant}$).

These hypotheses can be translated into statements about parameters within a probability model:

- (i) ' $p_1 = p_2$ ',
- (ii) ' $N \sim \text{Poi}(\lambda)$ for some $\lambda > 0$ ', i.e. : $p_n = \mathbb{P}(N = n) = \lambda^n \exp(-\lambda) / n!$
(within the general probability model $p_n \geq 0 \quad \forall n = 0, 1, \dots; \sum p_n = 1$),
- (iii) ' $\theta_1 \leq \theta_2$ ' and
- (iv) ' $\beta_1 = 0$ ' (assuming the linear model $E[Y | x] = \beta_0 + \beta_1 x$).

Definition 7.1 (Hypothesis Test): A hypothesis test is a procedure for deciding whether to accept a particular hypothesis as a reasonable simplifying assumption, or to reject it as unreasonable in the light of the data.

Definition 7.2 (Null Hypothesis): The null hypothesis H_0 is the default assumption we are considering making.

Definition 7.3 (Alternative Hypothesis): The alternative hypothesis H_1 is the alternative explanation(s) we are considering for the data.

Ordinarily, the null hypothesis is what we would assume to be true in the absence of data which suggests that it is not. Ordinarily, H_0 will explain the data in at least as simple a way as H_1 .

Definition 7.4 (Type I error): A type I error is made if H_0 is rejected when H_0 is true. In some situations this type of error is known as a false positive.

Definition 7.5 (Type II error): A type II error is made if H_0 is accepted when H_0 is false. This may also be termed a false negative.

Example 7.1: In the first example above (pandas) the null hypothesis is $H_0 : p_1 = p_2$.

The alternative hypothesis in the first example would usually be $H_1 : p_1 \neq p_2$, though it could also be (for example)

- (i) $H_1 : p_1 < p_2$,
- (ii) $H_1 : p_1 > p_2$, or
- (iii) $H_1 : p_1 - p_2 = \delta$ for some specified $\delta \neq 0$.

each of these alternative hypotheses makes a slightly different statement about the collection of situations which we believe are possible. A statistician needs to decide which type of hypothesis test is appropriate in any real situation.

7.2. Simple Hypothesis Tests

The simplest type of hypothesis testing occurs when the probability distribution giving rise to the data is specified completely under the null and alternative hypotheses.

Definition 7.6 (Simple Hypotheses): A simple hypothesis is of the form $H_k : \theta = \theta_k$, where θ is the parameter vector which parameterizes the probabilistic model for the data. A simple hypothesis specifies the precise value of the parameter vector (i.e. the probability distribution of the data is specified completely).

Definition 7.7 (Composite Hypotheses): A composite hypothesis is of the form $H_k : \theta \in \Omega_k$, i.e. the parameter θ lies in a specified subset Ω_k of the parameter space Ω_θ . This type of hypothesis specifies an entire collection of values for the parameter vector and so specifies a class of probabilistic models from which the data may have arisen.

Definition 7.8 (Simple Hypothesis Test): A simple hypothesis test tests a simple null hypothesis $H_0 : \theta = \theta_0$ against a simple alternative $H_1 : \theta = \theta_1$, where θ parameterizes the distribution of our experimental random variables $\mathbf{X} = X_1, X_2, \dots, X_n$.

Although simple hypothesis tests seem appealing, there are many situations in which a statistician cannot reduce the problem at hand to a clear dichotomy between two fully-specified models for the data-generating process.

There may be many seemingly sensible approaches to testing a given hypothesis. A reasonable criterion for choosing between them is to attempt to minimize the chance of making a mistake: incorrectly rejecting a true null hypothesis, or incorrectly accepting a false null hypothesis.

Definition 7.9 (Size): A test of size α is one which rejects the null hypothesis $H_0 : \theta = \theta_0$ in favour of the alternative $H_1 : \theta = \theta_1$ if and if

$$\mathbf{X} \in C_\alpha \text{ Where } \mathbb{P}(\mathbf{X} \in C_\alpha | \theta = \theta_0) = \alpha$$

for some subset C_α of the sample space S of $\mathbf{X}$.

The size, α , of a test is the probability of rejecting H_0 when H_0 is in fact true; i.e. size is the probability of type I error if H_0 is true. We want α to be small ($\alpha = 0.05$, say).

Definition 7.10 (Critical Region): The set C_α in Definition 7.9 is called the critical region or rejection region of the test.

Definition 7.11 (Power & Power Function): The power function of a test with critical region C_α is the function

$$\beta(\theta) = \mathbb{P}(\mathbf{X} \in C_\alpha | \theta),$$

and the power of a simple test is $\beta = \beta(\theta_1)$, i.e. the probability that we reject H_0 in favour of H_1 when H_1 is true.

Thus a simple test of power β has probability $1-\beta$ of a type II error occurring when H_1 is true. Clearly for a fixed size α of test, the larger the power β of a test the better.

However, there is an inevitable trade-off between small size and high power (as in a jury trial: the more careful one is not to convict an innocent defendant, the more likely one is to free a guilty one by mistake).

A hypothesis test typically uses a test statistic $T(\mathbf{X})$, whose distribution is known under H_0 , and such that extreme values of $T(\mathbf{X})$ are more compatible with H_1 than H_0 .

Many useful hypothesis tests have the following form:

Definition 7.12 (Simple Likelihood Ratio Test): A simple likelihood ratio test (SLRT) of $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$ rejects H_0 iff

$$\mathbf{X} \in C_\alpha^* \left\{ x \left| \frac{L(\theta_0; x)}{L(\theta_1; x)} \leq A_\alpha \right. \right\}$$

Where $L(\theta; x)$ is the likelihood of θ given the data x , and the number A_α is chosen so that the size of the test is α .

Exercise 7.2.1: Suppose that $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\theta, 1)$. Show that the likelihood ratio for testing $H_0 : \theta = 0$ against $H_1 : \theta = 1$ can be written

$$\lambda(x) = \exp \left[n \left(\bar{x} - \frac{1}{2} \right) \right].$$

Hence show that the corresponding SLRT of size α rejects H_0 when the test statistic $T(\mathbf{X}) = \bar{x}$ satisfies $T > \Phi^{-1}(1-\alpha)/\sqrt{2}$.

A number of points should be borne in mind:

- For a simple hypothesis test, both H_0 and H_1 are 'point hypotheses', each specifying a particular value for the parameter θ rather than a region of the parameter space.
- In practice, no hypothesis will be precisely true, so the whole foundation of classical hypothesis testing seems suspect! Actually, there is a tendency to overuse hypothesis testing: it is appropriate only when one really does wish to compare competing, well-defined hypotheses. In many problems the use of point estimation with an appropriate confidence interval is much easier to justify.
- Regarding likelihood as a measure of compatibility between data and model, an SLRT compares the compatibility of θ_0 and θ_1 with the observed data x , and accepts H_0 iff the ratio is sufficiently large.
- One reason for the importance of likelihood ratio tests is the following theorem, which shows that out of all tests of a given size, an SLRT (if one exists) is 'best' in a certain sense.

Theorem 7.1 (The Neyman-Pearson Lemma): Given random variables $X_1, X_2, \dots, X_n$, with joint density $f(x|\theta)$, the simple likelihood ratio test of a fixed size α for testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ is at least as powerful as any other test of the same size.

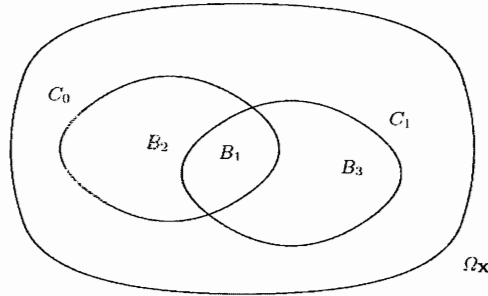
Proof: Fix the size of the test to be α .

Let A be a positive constant and C_0 a subset of the sample space satisfying

- (a) $\mathbb{P}(X \in C_0 | \theta = \theta_0) = \alpha$,
(b) $X \in C_0 \Leftrightarrow \frac{L(\theta_0; x)}{L(\theta_1; x)} = \frac{f(x|\theta_0)}{f(x|\theta_1)} \leq A$.

Suppose that there exists another test of size α , defined by the critical region C_1 , i.e.

Reject H_0 iff $x \in C_1$, where $\mathbb{P}(x \in C_1 | \theta = \theta_0) = \alpha$.



Let $B_1 = C_0 \cap C_1$, $B_2 = C_0 \cap C_1^c$, $B_3 = C_0^c \cap C_1$.

Note that $B_1 \cup B_2 = C_0$, $B_1 \cup B_3 = C_1$, and B_1, B_2 & B_3 are disjoint.

Let the power of the likelihood ratio test be $I_0 = \mathbb{P}(X \in C_0 | \theta = \theta_1)$,

and the power of the other test be $I_1 = \mathbb{P}(X \in C_1 | \theta = \theta_1)$.

We want to show that $I_0 - I_1 \geq 0$.

But

$$\begin{aligned} I_0 - I_1 &= \int_{C_0} f(x|\theta_1) dx - \int_{C_1} f(x|\theta_1) dx \\ &= \int_{B_1 \cup B_2} f(x|\theta_1) dx - \int_{B_1 \cup B_3} f(x|\theta_1) dx \\ &= \int_{B_2} f(x|\theta_1) dx - \int_{B_3} f(x|\theta_1) dx. \end{aligned}$$

Also $B_2 \subseteq C_0$, so $f(x|\theta_1) \geq A^{-1}f(x|\theta_0)$ for $x \in B_2$,

Similarly, $B_3 \subseteq C_0^c$, so $f(x|\theta_1) \leq A^{-1}f(x|\theta_0)$ for $x \in B_3$,

Therefore

$$\begin{aligned} I_0 - I_1 &\geq A^{-1} \left[\int_{B_2} f(x|\theta_0) dx - \int_{B_3} f(x|\theta_0) dx \right] \\ &= A^{-1} [\alpha - \alpha] = 0 \end{aligned}$$

as required.

7.3. Simple Null, Composite Alternative

Suppose that we wish to test the simple null hypothesis $H_0 : \theta = \theta_0$ against the composite alternative hypothesis $H_1 : \theta \in \Omega_1$.

The easiest way to investigate this is to imagine the collection of simple hypothesis tests with null hypothesis $H_0 : \theta = \theta_0$ and alternative $H_1 : \theta = \theta_1$, where $\theta_1 \in \Omega_1$. Then, for any given θ_1 , an SLRT is the most powerful test for a given size α . The only problem would be if different values of θ_1 result in different SLRTs.

Definition 7.13 (UMP Tests): A hypothesis test is called a uniformly most powerful test of $H_0 : \theta = \theta_0$ against $H_1 : \theta \in \Omega_1$, $\theta_1 \in \Omega_1$, if

- (i) There exists a critical region C_α corresponding to a test of size α not depending on θ_1 ,
- (ii) For all values of $\theta_1 \in \Omega_1$, the critical region C_α defines a most powerful test of $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$.

Exercise 7.3.1: Suppose that $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\theta, \sigma^2)$.

- (a.) Find the UMP test of $H_0 : \sigma^2 = 1$ against $H_1 : \sigma^2 > 1$.
- (b.) Find the UMP test of $H_0 : \sigma^2 = 1$ against $H_1 : \sigma^2 < 1$.
- (c.) Show that no UMP test of $H_0 : \sigma^2 = 1$ against $H_1 : \sigma^2 \neq 1$ exists.

There are several points to notice:

- If a UMP test exists, then it is clearly the appropriate test to use. It's the "best" test for comparing H_0 with any alternative consistent with H_1 .
- Often UMP tests don't exist! The requirements above are actually rather strong: in particular, requiring that a test is the most powerful test for a complicated class of alternative hypotheses limits the circumstances in which it is possible to specify such a test.
- Sufficient statistics are summaries of data which have the property that the distribution of the data, given the statistic, is conditionally independent of the data itself (i.e. $\mathbb{P}[X = x | T(x) = t, \theta] = \mathbb{P}[X = x | T(x) = t]$). A UMP test involves the data only via a likelihood ratio, so is a function of the sufficient statistics.
- As a result of the previous point, when a UMP test does exist, the critical region C_α often has a simple form, and is usually easily found once the distribution of the sufficient statistics have been determined (hence the importance of the χ^2 , t and F distributions).
- The above three examples illustrate how important is the form of alternative hypothesis being considered. The first two are one-sided alternatives whereas $H_1 : \sigma^2 \neq 1$ is a two-sided alternative hypothesis, since σ^2 could lie on either side of 1.

7.4. Composite Hypothesis Tests

The most general situation we'll consider is that in which the parameter space Ω is divided into two subsets: $\Omega = \Omega_0 \cup \Omega_1$, where $\Omega_0 \cap \Omega_1 = \emptyset$, and the hypotheses are $H_0 : \theta \in \Omega_0$, $H_1 : \theta \in \Omega_1$.

For example, one may want to test the null hypothesis that the data come from an exponential distribution against the alternative that the data come from a more general gamma distribution. In this case, the class of exponential distributions corresponds to the class of gamma distributions in which the shape parameter is required to be 1. Note that Ω_0 is a one-dimensional subspace of Ω , whilst Ω_1 is the remainder of the two-dimensional parameter space of the gamma distribution: here, as in many other cases, $\dim(\Omega_0) < \dim(\Omega_1) = \dim(\Omega)$.

One possible approach to this situation is to regard the maximum possible likelihood over $\theta \in \Omega_i$ as a measure of compatibility between the data and the hypothesis H_i ($i = 0, 1$).

It is convenient to define the following:

- $\hat{\theta}$ is the MLE of θ over the whole parameter space Ω ,
- $\hat{\theta}_0$ is the MLE of θ over Ω_0 , i.e. under the null hypothesis H_0 , and
- $\hat{\theta}_1$ is the MLE of θ over Ω_1 , i.e. under the alternative hypothesis H_1 .

Note that $\hat{\theta}$ must therefore be the same as either $\hat{\theta}_0$ or $\hat{\theta}_1$, because $\Omega = \Omega_0 \cup \Omega_1$.

One might consider using the likelihood ratio criterion $L(\hat{\theta}_1; x)/L(\hat{\theta}_0; x)$, by direct analogy with the SLRT. However, it is generally easier to use the related ratio $L(\hat{\theta}; x)/L(\hat{\theta}_0; x)$. Note that this agrees with the likelihood ratio criterion above when $\hat{\theta}_1$ has the higher likelihood and is otherwise equal to 1. This avoids the calculation of the MLE under the constraint that $\theta \in \Omega_1$ which will generally be somewhat more difficult than the calculation of the unconstrained MLE.

Definition 7.14 (Likelihood Ratio Test (LRT)): A likelihood ratio test rejects $H_0 : \hat{\theta} \in \Omega_0$ in favour of the alternative $H_1 : \hat{\theta} \in \Omega_1 = \Omega \setminus \Omega_0$ iff

$$\lambda(x) = \frac{L(\hat{\theta}; x)}{L(\hat{\theta}_0; x)} \geq \lambda,$$

Where $\hat{\theta}$ is the MLE of θ over the whole parameter space Ω , $\hat{\theta}_0$ is the MLE of θ over Ω_0 , and the value λ is fixed so that

$$\sup_{\theta \in \Omega_0} \mathbb{P}(\lambda(\mathbf{X}) \geq \lambda | \theta) = \alpha$$

Where α , the size of the test, is some chosen value.

It is possible to define an equivalent criterion in terms of the log likelihood and the log likelihood ratio test statistic:

$$r(x) = \ell(\hat{\theta}; x) - \ell(\hat{\theta}_0; x) \geq \lambda',$$

Where $\ell(\theta; x) = \log L(\theta; x)$, and λ' is chosen to give chosen size $\alpha = \sup_{\theta \in \Omega_0} \mathbb{P}(r(\mathbf{X}) \geq \lambda' | \theta)$. As is often the case, taking logarithms can simplify computations. Equation 7.2 is often easier to work with than Equation 7.1—see the exercises and problems.

Put Your Own Notes

As ever, the size α is typically chosen by convention to be 0.05 or 0.01.

High values of the test statistic $\lambda(x)$, or equivalently of $r(x)$, are taken as evidence against the null hypothesis H_0 . If $\lambda(x)=1$ or equivalently $r(x)=0$ then the null hypothesis explains the observed data at least as well as the alternative hypothesis.

A note on terminology: The test given in Definition 7.14 is sometimes referred to as a generalized likelihood ratio test, and Equation 7.1 a generalized likelihood ratio test statistic. When this terminology is used the simple likelihood ratio test defined here in definition 7.12 may be referred to as a likelihood ratio test.

These abstract tests may seem far removed from real statistical problems. However, as the following exercises demonstrate, standard widely-used procedures can be readily obtained as particular cases of the type of procedure introduced above. An understanding of the general case can be invaluable when trying to interpret or apply specific examples.

Exercise 7.4.1 (Paired t -test): Suppose that $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, and Let $\bar{X} = \sum X_i / n$, $S^2 = \sum (X_i - \bar{X})^2 / (n-1)$. What is the distribution of $T = \bar{X} / (S / \sqrt{n})$?

Is the test based on rejecting $H_0 : \mu = 0$ for large T a likelihood ratio test?

Assuming that the observed differences in diastolic blood pressure (after-before) are iid and Normally distributed with mean δ_D , use the captopril data (Table 7.1) to test the null hypothesis $H_0 : \delta_D = 0$ against the alternative hypothesis $H_1 : \delta_D \neq 0$. This procedure is known as the paired t test.

Patient Number	Systolic			Diastolic		
	Before	After	Change	Before	After	Change
1	210	201	-9	130	125	-5
2	169	165	-4	122	121	-1
3	187	166	-21	124	121	-3
4	160	157	-3	104	106	2
5	167	147	-20	112	101	-11
6	176	145	-31	101	85	-16
7	185	168	-17	121	98	-23
8	206	180	-26	124	105	-19
9	173	147	-26	115	103	-12
10	146	136	-10	102	98	-4
11	174	151	-23	98	90	-8
12	201	168	-33	119	98	-21
13	198	179	-19	106	110	4
14	148	129	-19	107	103	-4
15	154	131	-23	100	82	-18

Table 7.1: Supine systolic and diastolic blood pressures of 15 patients with moderate hypertension (high blood pressure), immediately before and 2 hours after taking 25 mg of the drug captopril.

Definition 7.15 (Chi-squared Goodness of Fit Statistic):

$$X^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i},$$

Where o_i is the observed count in the i th category and e_i is the corresponding expected count under the null hypothesis, is called the χ^2 goodness-of-fit statistic.

Under H_0 , X^2 has approximately a χ^2 distribution with number of degrees of freedom being (number of categories) - 1 - (number of parameters estimated under H_0).

This approximation works well provided all the expected counts are reasonably large (in practice, it's often employed when the expected number of counts are all at least 5 and works tolerably well even here).

This χ^2 test was suggested by Karl Pearson before the theory of hypothesis testing was fully developed.

Test of Independence in a Contingency Table: The same test statistic can also be used to test for independence of variables in a contingency table. In this case, the row and column totals are fixed in addition to the grand total, hence we lose $1 + (r-1) + (c-1)$ degrees of freedom, leaving $rc - r - c + 1 = (r-1)(c-1)$ degrees of freedom.

Example 7.2: Use the following 3×3 contingency table to test at the 0.01 level of significance whether a person's ability in mathematics is independent of her/his interest in statistics.

Ability in Maths

		Low	Average	High	Total
Interest in Stats	Low	63	42	15	120
	Average	58	61	31	150
	High	14	47	29	90
	Totals	135	150	75	360

H_0 : Ability in maths and interest in statistics are independent.

Versus H_1 : Ability in maths and interest in statistics are not independent.

Decision rule: reject H_0 if $X^2 > \chi_{0.01,4}^2 = 13.277$ where

$$X^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(\text{observed}_{ij} - \text{expected}_{ij})^2}{\text{expected}_{ij}}$$

Under H_0 , i.e. independence, the expected frequencies are given by the product of corresponding marginal estimated probabilities times the total number of individuals. For example for the first row these are

$$\left(\frac{120}{360}\right)\left(\frac{135}{360}\right)360 = 45, \left(\frac{120}{360}\right)\left(\frac{150}{360}\right)360 = 50, \text{ and } 120 - 45 - 50 = 25. \text{ Thus}$$

$$X^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(63-45)^2}{45} + \frac{(42-50)^2}{50} + \dots + \frac{(29-18.75)^2}{18.75} = 32.14$$

Since $32.14 > 13.277$ the null hypothesis must be rejected.

CHAPTER 8

ELEMENTS OF BAYESIAN INFERENCE

Thus far we have considered only the so-called classical or frequentist approach to statistical inference. In the classical paradigm, unknown parameters are treated as fixed but unknown constants which are to be estimated. Probabilistic statements are made only about “true” random variables and observed data is assumed to correspond to a sampled set of realizations of random variables. Bayesian inference provides an alternative approach

8.1. Introduction

In the Bayesian approach to inference, parameters are treated as random variables and hence have a probability distribution.

Prior information about θ is combined with information from sample data to estimate the distribution of θ . This distribution contains all the available information about θ so should be used for making estimates or inferences.

We have prior information about θ given by the prior distribution, $p(\theta)$, and information from sample data given by the likelihood $L(\theta; x) = f(x; \theta)$. By Bayes Theorem the conditional distribution of θ given $X = x$ is.

$$q(\theta|x) = \frac{f(x; \theta)p(\theta)}{h(x)} = \frac{L(\theta; x)p(\theta)}{h(x)} \propto L(\theta; x)p(\theta)$$

Where $h(x)$ is the marginal distribution of x . We call $q(\theta|x)$ the posterior distribution.

Actually, a Bayesian would most probably have written:

$$q(\theta|x) = \frac{f(x|\theta)p(\theta)}{h(x)}$$

There is no need to distinguish between parameters and random variables in notation and it's perfectly reasonable to condition upon the parameters within the Bayesian framework

Note: A Bayesian statistician does not necessarily believe that all parameter values are classical random variables. Viewing probabilistic statements as quantifications of uncertainty without explicit reference to relative frequencies of occurrence allows their use much more widely. In the subjective Bayesian framework, probabilities are quantifications of personal belief in a statement.

Prior Distribution: The prior distribution $p(\theta)$ quantifies information about θ prior to the gathering of the current data.

Sometimes $p(\theta)$ can be constructed on the basis of past data. More commonly, $p(\theta)$ must be based upon an expert's experience and personal judgments. The following three examples give some simplified examples of the procedures by which such distributions can be obtained.

Example 8.1: Suppose that the proportions θ of defective items in a large manufactured lot is unknown.

The prior distribution assigned to θ might be $U(0, 1)$, i.e.

$$p(\theta) = \begin{cases} 1 & \text{for } 0 < \theta < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Example 8.2: Suppose that the lifetimes of lamps of a certain type are to be observed. Let X be the lifetime of any lamp and let $X \sim \text{Exp}(\beta)$, where β is unknown.

On the basis of previous experience the prior distribution of β is taken as a gamma distribution with mean 0.0002 and standard deviation 0.0001, i.e.

$$p(\beta) = \begin{cases} \frac{20000^4}{3!} \beta^3 e^{-20000\beta}, & \beta > 0 \\ 0 & \text{otherwise} \end{cases}$$

Example 8.3: A medical researcher was questioned about, the proportion of asthma sufferers who would be helped by a new drug.

She thought that

$$P[\theta > 0.3] = P[\theta < 0.3]$$

i.e. that the median $\theta_{0.5} = 0.3$

Similarly, she thought that

$$\theta_{0.25} = 0.2 \text{ and } \theta_{0.75} = 0.45$$

From tables giving quantiles of beta distributions, the researcher's opinion could be represented by Beta ($\alpha = 2, \beta = 4$) for which

$$\theta_{0.25} = 0.194, \quad \theta_{0.5} = 0.314 \text{ and } \theta_{0.75} = 0.454$$

Note that fusing information from different sources and extracting the knowledge of domain experts in order to produce prior distributions which genuinely encode the state of knowledge prior to the gathering of a first data set is a difficult problem. A large amount of research has been done on the area which is termed prior elicitation. If one conducts a series of experiments concerned with the same parameter then the situation is improved a little as we shall see the posterior distribution $p(\theta | x_1)$ can be used as the prior distribution when the next data set x_2 , say, is obtained and this procedure can be applied iteratively as we learn progressively more about the parameter.

8.2. Parameter Estimation

Suppose we wish to estimate a parameter θ . We define a loss function.

$$L_s(\theta, \hat{\theta})$$

Which measures the loss which would be suffered as a result of using $\hat{\theta}$ as an estimator when the true value is θ . This is a formal way of encoding how bad various types of possible error in the estimation are to us. For instance, it can encode the difference between type I and type II errors if we are interested in estimating a binary indicator of disease status.

The Bayes estimator minimizes the expected loss:

$$E[L_s(\theta, \hat{\theta}) | x] = \int_{-\infty}^{\infty} L_s(\theta, \hat{\theta}) q(\theta | x) d\theta$$

for the observed value of x .

The form of the Bayes estimator depends on the loss function that is used and the prior that is assigned to θ .

For example, if the loss is the absolute error

$$L_s(\theta, \hat{\theta}) = |\theta - \hat{\theta}|$$

then the Bayes estimator $\hat{\theta}_B(x)$ is the median of the posterior distribution.

For other loss functions the minimum might have to be numerically estimated.

Example 8.2.1: A commonly used loss function is the squared error loss function

$$L_s(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$$

Show that the corresponding Bayes estimator $\hat{\theta}_B$ is equal to the mean of the posterior distribution.

$$\hat{\theta}_B(x) = E[\theta | x]$$

What is the minimum expected loss?

Example 8.4 (Continuing example 8.1): Suppose that a random sample of n items is taken from the lot of manufactured items. Let

$$X_i = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ item is defective} \\ 0 & \text{otherwise} \end{cases}$$

Then $X_1, \dots, X_n$ is a sequence of Bernoulli trials with parameter θ . The pdf of each X_i is

$$f(x | \theta) = \begin{cases} \theta^x (1 - \theta)^{1-x} & \text{for } x = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

Then the joint pdf of $X_1, \dots, X_n$ is

$$f_n(x | \theta) = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}$$

Since the prior pdf $p(\theta)$ is uniform it follows that

$$f_n(x | \theta) p(\theta) = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}, \quad 0 \leq \theta \leq 1$$

This is proportional to a beta distribution with parameters $\alpha = y + 1$ and $\beta = n - y + 1$, where $y = \sum x_i$.

Therefore the posterior has pdf

$$q(\theta | x) = \frac{\Gamma(n+2)}{\Gamma(y+1)\Gamma(n-y+1)} \theta^y (1 - \theta)^{n-y}, \quad 0 \leq \theta \leq 1.$$

Example 8.5: (Example 8.2 continued) Suppose that the lifetime $X_1, \dots, X_n$ of a random sample of n lamps are recorded. The pdf of each x_i is

$$f(x, \beta) = \begin{cases} \beta e^{-\beta x_i} & x > 0, \\ 0 & \text{otherwise} \end{cases}$$

The joint pdf of $x_1, \dots, x_n | \beta$ is

$$f(x | \beta) = \beta^n e^{-\beta y}, \text{ where } y = \sum_{i=1}^n x_i.$$

with a gamma specified for $p(\beta)$ we have

$$f(x | \beta) p(\beta) \propto \beta^{n+3} e^{-(y+20000)\beta}$$

Where a factor that is constant w. r. t. β has been omitted. The RHS is proportional to a Gamma $(n+4, y+20000)$, hence

$$q(\beta | x) = \frac{(y+20000)^{n+4}}{(n+3)!} \beta^{n+3} e^{-(y+20000)\beta}$$

8.3. Conjugate Prior Distributions

A conjugate prior distribution when combined with the likelihood function produces a posterior distribution in the same family as the prior.

If we find a conjugate prior distribution which adequately fits our prior beliefs regarding θ , we should use it because it will simplify computations considerably. However, one should not employ a conjugate prior distribution for computational convenience if it does not represent those prior beliefs reasonably closely.

Example 8.6: (Sampling from a Bernoulli Distribution). Suppose $X_1, \dots, X_n$ are a random sample from $\text{Ber}(\theta)$, $0 < \theta < 1$. Let $p(\theta)$ be Beta (α, β) .

Then $q(\theta | x)$ is Beta $\left(\alpha + \sum_{i=1}^n x_i, \beta + n - \sum_{i=1}^n x_i\right)$.

The proof of this claim is analogous to Example 8.4 (note $U[0,1] \equiv \text{Beta}(1,1)$).

Exercise 8.3.1 (Sampling from a Poisson distribution). Suppose $X_1, \dots, X_n$ are a random sample from $\text{Poi}(\theta)$. Let $p(\theta)$ be Gamma (α, β) .

Show that the posterior density, $q(\theta | x)$, is

$$\text{Gamma} \left(\alpha + \sum_{i=1}^n x_i, \beta + n \right)$$

Example 8.7: (Sampling from a Normal Distribution with known σ^2). Suppose $X_1, \dots, X_n$ are a random sample from $N(\theta, \sigma^2)$. with σ^2 known. Let $p(\theta)$ be $N(\phi, \tau^2)$.

Then $q(\theta | x)$ is

$$N \left(\frac{\phi \sigma^2 + n \bar{x} \tau^2}{\sigma^2 + n \tau^2}, \left(\frac{\sigma^2 + n \tau^2}{\sigma^2 \tau^2} \right)^{-1} \right)$$

Proof:

$$q(\theta|x) \propto p(\theta)L(\theta;x)$$

$$= (2\pi\tau^2)^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\left(\frac{\theta-\phi}{\tau}\right)^2\right\}$$

$$\times \prod_{i=1}^n (2\pi\sigma^2)^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\left(\frac{x_i-\theta}{\sigma}\right)^2\right\}$$

$$\therefore q(\theta|x) \propto \exp\left\{-\frac{1}{2}\left[\theta^2\left(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\right) - 2\theta\left(\frac{\phi}{\tau^2} + \frac{\sum x_i}{\sigma^2}\right)\right]\right\}$$

$$= \exp\left\{-\frac{\sigma^2 + n\tau^2}{2\sigma^2\tau^2}\left[\theta - \frac{\phi\sigma^2 + n\bar{x}\tau^2}{\sigma^2 + n\tau^2}\right]^2 + \text{constant}\right\}$$

i.e. $q(\theta|x)$ is the pdf of $N\left(\frac{\phi\sigma^2 + n\bar{x}\tau^2}{\sigma^2 + n\tau^2}, \left(\frac{\sigma^2 + n\tau^2}{\sigma^2\tau^2}\right)^{-1}\right)$ as required.

Precision: Note that the posterior variance

$$\left(\frac{\sigma^2 + n\tau^2}{\sigma^2\tau^2}\right)^{-1} = \left(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\right)^{-1}$$

i.e. the reciprocal of the sum of the reciprocals of the prior variance and the variance of the sample mean, respectively.

Because of this reciprocal relationship

$$\text{Precision} = \frac{1}{\text{variance}}$$

Is sometimes quoted instead of the variance.

Posterior Mean: The posterior mean

$$\frac{\phi\sigma^2 + n\bar{x}\tau^2}{\sigma^2 + n\tau^2} = \frac{\frac{\phi}{\tau^2} + \frac{\bar{x}}{\sigma^2/n}}{\frac{1}{\tau^2} + \frac{1}{\sigma^2/n}}$$

i.e. a weighted average of the prior mean ϕ and the sample mean $\bar{x}$, with weights proportional to the prior precision and the precision of the sample mean.

This type of relationship holds for several sampling distributions when a conjugate prior is used.

Example 8.8: (Sampling from an Exponential Distribution). Suppose $X_1, \dots, X_n$ are a random sample from $\text{Exp}(\theta)$. Let $p(\theta)$ be Gamma (α, β) .

Then $q(\theta|x)$ is

$$\text{Gamma}\left(\alpha + n, \beta + \sum_{i=1}^n x_i\right)$$

See example 8.5 for a proof of this claim.

CHAPTER 9

MARKOV CHAINS

Stochastic Process: The collection of R.V. $\{X_t; t \in T\}$ defined on some probability space (Ω, F, P) is called stochastic process.

Here, the set T is called index set & the values that R.V. X_t takes is called state space and is denoted by S .

We will study the case when T is discrete (i.e. a countable set) and state space is also discrete.

Example: $\{X_t; t \in \mathbb{R}\}$ where X_t is 1 if head & 0 if tail & we toss the coin daily.

9.1. Discrete Time Markov Chain

A sequence $(X_n)_{n \in \mathbb{N}}$ of R.V. with discrete state-space S is called DTMC if

$$P(X_{n+1} = j | X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_1 = i_1) \\ = P(X_{n+1} = j | X_n = i_n) \quad \forall i_1, \dots, i_n, j \in S$$

i.e. the future value depends only on the present value not on the past history.

Examples: Suppose that a coin is tossed repeatedly and let

X_n := "number of heads in n tosses".

Definitions

- (i) **Transition Probability:** $P_{ij} := P(X_{n+1} = j | X_n = i)$ i.e. probability of going from state i to j in one step.
- (ii) **Transition Probability Matrix:** Given the one step transition probabilities, we can write it in the matrix form as:

$$P = (P_{ij})_{i,j \in S} = \begin{pmatrix} P_{00} & P_{01} & P_{02} & \dots \\ P_{10} & P_{11} & P_{12} & \\ P_{20} & P_{21} & P_{22} & \\ \vdots & & & \ddots \end{pmatrix}$$

Note that each $P_{ij} \geq 0 \quad \forall i, j$ and each row sum has to be 1.

The matrix whose each row sum is 1 is called stochastic matrix.

And if both row & column sum is 1, then it is called a doubly stochastic matrix.

- (iii) **Initial Distribution:** The probability distribution $\pi := (\pi_i)_{i \in S}$

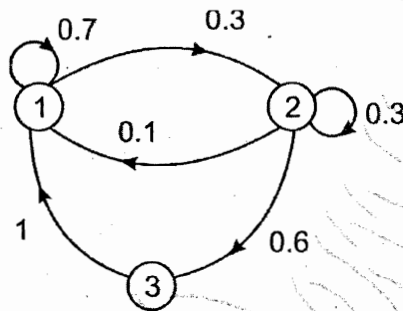
$\pi_i := P(X_0 = i)$ is called initial distribution. It basically gives you the probability that the chain starts at i^{th} state initially.

i.e. Probability of system starting at state i is π_i

- (iv) **State Transition Diagram:** Given the matrix P , we can draw a state transition diagram & vice-versa. The diagram consists of circles showing the state and arcs from one circle to other showing whether it is possible to go from that state to the other.

Example:

$$P = \begin{pmatrix} 0.7 & 0.3 & 0 \\ 0.1 & 0.3 & 0.6 \\ 1 & 0 & 0 \end{pmatrix}$$



- (v) **M-Step Transition Probabilities:** We can define m step transition probabilities that gives you the probability of going from i^{th} state to j^{th} state in m steps. i.e.

$$P_{ij}^{(m)} = P(X_{n+m} = j | X_n = i) \quad \forall i, j \in S$$

So, we can define a m -step transition matrix denoted by $P^{(m)}$

$$\text{as } P^{(m)} = P^m$$

- (vi) **Accessibility:** State j is said to be accessible from the state i if

$$P_{ij}^{(n)} > 0 \text{ for some } n.$$

i.e. if we can go from i to j in n steps for some n .

- (vii) **Communicating States:** Two states i and j are said to be communicating if $i \rightarrow j$ and $j \rightarrow i$.

i.e. if it is possible to go from i to j in n steps and then back to i in m steps for some m, n needn't be same.

- (viii) **Class of a State i :**

$$C(i) = \{j \in S : i \leftrightarrow j\}, i \in S$$

$$C(i) = \{\text{Set of all states that communicate with } i\}$$

$$i \in C(i) [\because i \text{ Communicate with } i \text{ in zero steps}]$$

Closed Communicating Class: A class is said to be closed if the exit from that class is not possible i.e. once you entered that class, you get trapped there. A closed communicating class consists of recurrent states only.

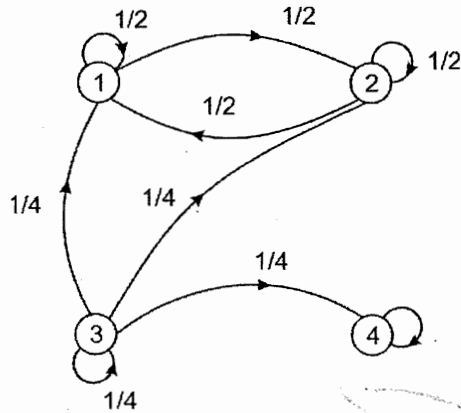
- (a) **Irreducible and Reducible Chain:** A Markov Chain is said to be irreducible if every state communicate with every other state.

i.e. $C(i) = S$ otherwise chain is said to be reducible.

Put Your Own Notes

Example:

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



$C(1) = \{1, 2\}$ [$\because$ 1 to 2 and 2 to 1 is possible]

$C(2) = C(1)$

$C(3) = \{3\}$ [$\because$ $3 \rightarrow 1, 3 \rightarrow 2, 3 \rightarrow 4$ is possible, but not back]

$C(4) = \{4\}$

Since there are 3 classes, $\therefore$ chain is reducible.

(ix) **Period of a state i :** Let $i \in S$

$$\lambda(i) = d(i) = \gcd \left\{ n \geq 1 \mid P_{ii}^{(n)} > 0 \right\}$$

i.e.

Steps to calculate the period of any state i .

(a) first collect the number of steps in

Which $i \rightarrow i$ is possible.

i.e. $i \rightarrow i$ 2 steps

$i \rightarrow i$ 5 steps etc.

(b) Then take g.c.d. of these no's and this g.c.d gives you the period of state i .

Example: In the above example

$$\lambda(1) = \gcd \{1, 2, 3, 4, \dots\} = 1$$

$$\lambda(2) = \lambda(1) = 1$$

$$\lambda(3) = \gcd \{1, 2, 3, \dots\} = 1$$

$$\lambda(4) = 1$$

Result: If $i \leftrightarrow j$, then $\lambda(i) = \lambda(j)$.

Put Your Own Notes

- (x) **Aperiodic State & Chain:** State i is said to be aperiodic if

$$\lambda(i) = 1$$

If all states are aperiodic, then we say chain is aperiodic.

- (xi) **First Visit Probability:**

$$f_{jk}^{(m)} = P(X_m = k, X_p \neq k \quad \forall p < m \mid X_0 = j)$$

i.e. starting from j , going to k in m steps but cannot visit k in less than m steps. It must be in m th step only.

$$f_{ii} = \sum_{n=1}^{\infty} f_{ii}^{(n)} \text{ denotes the probability of ultimately returning to state } i.$$

- (xii) **Recurrent & Transient States:**

If $f_{ii} = 1$, then i is a recurrent state if $f_{ii} < 1$ then state i is transient state.

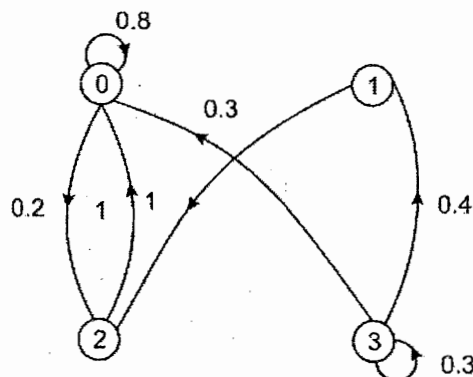
(transient mean that is non-zero probability that it will never return to it)

Note: Recurrent means, returning to that state is sure.

Transient means, return to state j is uncertain. After some finite number of steps, the chain will not come to that state.

Example:

$$P = \begin{pmatrix} 0.8 & 0 & 0.2 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0 & 0.3 \end{pmatrix}$$



$$C(0) = \{0, 2\} = (12)$$

$$C(1) = \{1\}$$

$$C(3) = \{3\}$$

$$f_{00}^{(1)} = 0.8$$

$$f_{00}^{(2)} = (0.2)(1) = 0.2$$

$$\therefore f_{00} = 0.8 + 0.2 = 1$$

$$f_{22} \text{ is also } 1 \text{ } [\because 0 \leftrightarrow 2]$$

$$f_{11}^{(1)} = 0 = f_{11}^{(2)} = f_{11}^{(3)} = \dots \therefore f_{11} = 0 < 1$$

$$f_{33}^{(1)} = 0.3, f_{33}^{(2)} = 0 \quad [\therefore \text{first visit}]$$

$$\therefore f_{33} = 0.3 [< 1]$$

$\therefore 0, 2$ are recurrent states and $1, 3$ are transient states.

Results:

- (a) A finite Markov chain cannot have all transient states.
- (b) A finite irreducible chain has all states recurrent (+ve recurrent).
- (c) If $i \leftrightarrow j$, then

(a) If i is recurrent then j is recurrent.

(b) If i is transient then j is transient.

- (xiii) Mean Recurrence time i.e. expected time of return to state i starting from state i .

$$\mu_i := \begin{cases} \sum_{n=1}^{\infty} n f_{ii}^{(n)} & \text{if } i \text{ is recurrent} \\ \infty & \text{if } i \text{ is transient} \end{cases}$$

Let i is recurrent state then we define

If $\mu_i < \infty$ then i is positive recurrent

If $\mu_i = \infty$ then i is null recurrent.

- (xiv) **Limiting Probabilities:**

$\lim_{n \rightarrow \infty} P_{ij}^{(n)}$ i.e. starting from i what is the probability of coming to j in n steps where n tends to ∞ .

(a) If j is transient $\lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0 \quad \forall i \in S$

(b) If i is recurrent & $\lambda(i) = 1$

$$\text{then } \lim_{n \rightarrow \infty} P_{ii}^{(n)} = \frac{1}{\mu_i}$$

In particular, if i is null recurrent, then

$$\lim_{n \rightarrow \infty} P_{ii}^{(n)} = 0$$

Result:

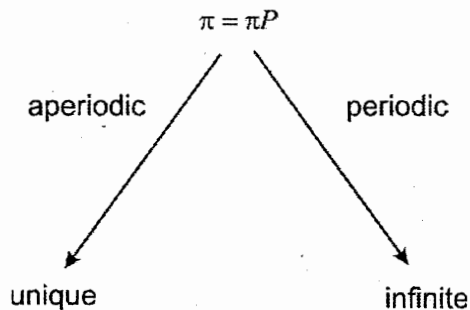
- (a) If $\{X_n; n \geq 0\}$ is an irreducible chain whose all states are positive recurrent & aperiodic then

$$\lim_{n \rightarrow \infty} P_{ij}^{(n)} = \frac{1}{\mu_j} = \pi_j \quad (\text{independent of } i) \text{ where } \pi_j \text{ is stationary distribution.}$$

- (b) If chain is periodic or state is periodic, then limiting probability doesn't exist.

(xv) **Stationary Distribution:** $\pi = \pi P$

Finite irreducible chain



But if aperiodic, irreducible & +ve recurrent the unique.

Note: It mayn't exist but if it exists then either unique or infinite. (we can't have say 5 stationary distributions).

Note: if stationary distribution is unique

i.e. chain is aperiodic, irreducible & all states +ve recurrent then

$$\lim_{n \rightarrow \infty} P_{ij}^{(n)} = \pi_j \text{ independent of } i$$

Put Your Own Notes

CHAPTER 10

MISCELLANEOUS

10.1. Reliability Analysis

Note: In these problems, we are dealing with life; $\therefore$ our possible values will always be from 0.

Survival Function and Hazard Rate Function: Let X denotes the life time of some device. Then

(i) Survival function/Reliability function

$$S(x) = P(X > x) = 1 - P(X \leq x) = 1 - F(x)$$

i.e. the probability that device is still working after time x .

(ii) Hazard Rate Function/Failure Rate Function

$$h(t) = \frac{d}{dt}(-\log(1 - F(t))) = \frac{f(t)}{1 - F(t)}$$

$$= -\frac{d}{dt}(\log S(t))$$

where $f(t)$ is PDF and $F(t)$ is CDF of system.

(iii) Expected Lifetime of the System

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

where $f(x)$ is PDF of system

Type of Questions:

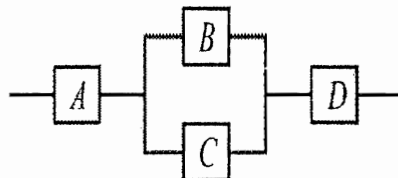
Type-I: When reliability is given to be a constant

Consider the system with components A, B, C, D whose reliability is $\frac{1}{2}$ for each component. Find the reliability of the system.

Solution: $P(X > x) = P(A \cap (B \cup C) \cap D)$

(i.e. A, D works, atleast one of $B \& C$ works, then system is working)

Since each device is working independently



$$= P(A)P(B \cup C)P(D)$$

$$= P(A)[P(B) + P(C) - P(B \cap C)]P(D)$$

$$= P(A)[P(B) + P(C) - P(B)P(C)]P(D)$$

$$= \frac{1}{2} \times \left[\frac{1}{2} + \frac{1}{2} - \frac{1}{2} \times \frac{1}{2} \right] \times \frac{1}{2}$$

$$= \frac{3}{16}$$

Type-II: (When reliability is some random variable & mostly exponential random variable and you have to find survival function or expected life)

Note: X is said to be exponential R.V. with parameter λ

$$f(x) = \begin{cases} \lambda e^{-\lambda x}; & x > 0 \\ 0; & \text{otherwise} \end{cases}$$

$$\text{i.e. } F(x) = P(X \leq x) = \begin{cases} 0; & x < 0 \\ 1 - e^{-\lambda x}; & 0 \leq x \end{cases}$$

$$\Rightarrow P(X > x) = 1 - P(X \leq x)$$

$$= \begin{cases} 1; & x < 0 \\ e^{-\lambda x}; & x \geq 0 \end{cases}$$

$$\& \text{ mean } E(X) = \int_0^{\infty} x f(x) dx = \int_0^{\infty} x (\lambda e^{-\lambda x}) dx = \frac{1}{\lambda}$$

Problem (June 2013) (Section B): Consider a parallel system with two components each having exponential distribution with mean one and life time of both components are independent. Find expected life of the system?

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{3}{2}$

(d) 2

Solution: To find expected lifetime of X , we need PDF of X .

(a) Survival Function

$$S(t) = P(X > t)$$

$$= P(X_1 > t) P(X_2 \leq t)$$

[$\because$ Independent]

$$+ P(X_1 \leq t) P(X_2 > t)$$

$$+ P(X_1 \geq t) P(X_2 \geq t)$$

(i.e. atleast one of them is working)

or

$$S(t) = P(X > t) = 1 - P(X \leq t)$$

$$= 1 - P(X_1 \leq t) P(X_2 \leq t)$$

i.e. $1 - P$ (system not working)

i.e. $1 - P$ (both not working)

$$\therefore S(t) = 1 - (1 - e^{-t})(1 - e^{-t})$$

$$= 1 - (1 - 2e^{-t} + e^{-2t})$$

$$P(X > t) = 2e^{-t} - e^{-2t}$$

$$\therefore F(x) = 1 - 2e^{-x} + e^{-2x}$$

$$\Rightarrow f(x) = 2e^{-x} - 2e^{-2x}$$

$$\therefore \text{Expected life time} = \int_0^{\infty} x (2e^{-x} - 2e^{-2x}) dx$$

$$= 2 - \frac{1}{2} = \frac{3}{2}$$

Note: Once we know $S(t)$, we can find hazard rate function also.

Type-III Comparing Two Systems

June 2012 (Part B)

The hazard rate of two life times variables T_1 and T_2 with respective CDF's $F_1(t)$ and $F_2(t)$ and PDF's $f_1(t)$ and $f_2(t)$, are $h_2(t) = 4t^3$ and $h_1(t) = 3t^2$, $t > 0$, then

- (a) $F_1(t) \geq F_2(t) \quad \forall t > 0$
- (b) $F_1(t) < F_2(t) \quad \forall t > 1$
- (c) $E(T_1) < E(T_2)$
- (d) $f_1(t) < f_2(t) \quad \forall t > 0$

Ans: b

Solution: $h_1(t) = -\frac{d}{dt} \log(S_1(t)) = 3t^2$

$$\Rightarrow -\log(S_1(t)) = t^3$$

$$\Rightarrow S_1(t) = e^{-t^3}$$

$$\Rightarrow F_1(t) = 1 - e^{-t^3}$$

Similarly, $F_2(t) = 1 - e^{-t^4}$

$$f_1(t) = 3t^2 e^{-t^3}$$

$$f_2(t) = 4t^3 e^{-t^4}$$

$$\forall t > 1, t^3 < t^4$$

$$-t^3 > -t^4$$

$$\Rightarrow e^{-t^3} > e^{-t^4}$$

[$\because e^x$ is strictly increasing]

$$\Rightarrow -e^{-t^3} < -e^{-t^4}$$

$$\Rightarrow 1 - e^{-t^3} < 1 - e^{-t^4}$$

$$\Rightarrow F_1(t) < F_2(t)$$

$\therefore$ Option (b) is correct

10.2. Quadratic Forms

Definition: Let $\{X_1, X_2, \dots, X_n\}$ be n (non random) variables. A quadratic form Q is, by definition, an expression such as:

$$Q = \sum_{ij} a_{ij} X_i X_j$$

Where (the coefficients of the form) are real numbers.

So, a quadratic form is a second degree, homogenous (no constant term) polynomial in the X_i .

Matrix Notation:

(i) A denotes the matrix whose general term is $[a_{ij}]$.

(ii) A quadratic form Q is then defined by the matrix equation:

$$Q = X' A X$$

Where $X = (X_1, X_2, \dots, X_n)$ is random vector.

Note: The matrix A is assumed to be symmetric matrix.

Expectation of a Quadratic Form: Let

- (i) X be a random vector with mean μ and covariance matrix Σ .
- (ii) $Q = X'AX$ be a quadratic form.

Then $E[X'AX] = \text{tr}(A\Sigma) + \mu'A\mu$

Note: The above formula is independent of the distribution of X .

Quadratic Forms in a Multivariate Normal Vector and The Chi-Square Distribution:

- (i) X is a multivariate normal vector then we write $X \sim N(\mu, \Sigma)$ where μ is mean vector and Σ is variance-covariance matrix.
- (ii) Let $\{X_1, X_2, \dots, X_n\}$ be n standard normal independent variables:
 $X_i \sim N(0, 1)$ for all i .

Then Chi-square distribution with n degrees of freedom is the distribution of the variable X^2 defined as:

$$X^2 = \sum_i X_i^2$$

and is usually denoted by $X^2 \sim \chi_n^2$.

Observe that it is a special quadratic form with A same as Identity matrix of order n .

Question: Do there exist some other matrix P such that $X'PX$ follows chi-square distribution?

Results:

- (i) If $X \sim N(0, I_n)$ then $Q = X'PX \sim \chi_r^2$ if and only if P is idempotent and rank (P) is r .
- (ii) If $X \sim N(\mu, \Sigma)$ then $Q = X'PX \sim \chi_r^2$ (non central) if and only if both conditions are satisfied
 - (a) $P = P \Sigma P$
 - (b) $\text{rank}(P) = r$.

Further, non centrality parameter is $\mu'P\mu$.

Independence of two quadratic forms in a multivariate normal vector

- (iii) Two Linear forms $A'X$ and $B'X$ in a multinormal vector X with covariance matrix Σ are independent if and only if $A'\Sigma B = 0$.
- (iv) **Craig's theorem:** Let $X \sim N_p(\mu, \Sigma)$, and let $X'AX$ and $X'BX$ be two (Chi-square distributed) quadratic forms, then these two forms are independent if and only if $A'\Sigma B = 0$.

Note: It is not assumed that the Chi-square distributions of the quadratic forms are central or have the same numbers of degrees of freedom

Put Your Own Notes

ASSIGNMENT SHEET-1

1. Classify the states of the Markov chains with the following transition probabilities

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix} \quad P = \begin{pmatrix} 0 & 0 & 1/3 & 2/3 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

2. Classify the states of the Markov chains with the following transition probabilities

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 1/3 & 0 \\ 0 & 0 & 2/3 & 1/3 & 0 \\ 1/3 & 1/3 & 0 & 0 & 1/3 \end{pmatrix}$$

3. Let $\{X_n; n \geq 0\}$ be a Markov chain with states 0, 1, 2 and with transition probability matrix

& initial distribution : $= \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$

$$P = \begin{pmatrix} 0.75 & 0.25 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0.75 & 0.25 \end{pmatrix}$$

Find

(i) $P(X_2 = 2, X_1 = 1, X_0 = 2)$

(a.) 0.625

(b.) 0.25

(c.) 0.75

(d.) 0.5

(ii) $P(X_2 = 2 | X_1 = 1)$

(a.) 0.5

(b.) 0.75

(c.) 0

(d.) 0.25

(iii) $P(X_2 = 2, X_1 = 1 | X_0 = 2)$

(a.) $\frac{3}{16}$

(b.) $\frac{2}{16}$

(c.) $\frac{1}{16}$

(d.) $\frac{5}{16}$

4. Let $\{X_n; n \geq 1\}$ be a Markov chain with state space $S = \{0, 1\}$, initial distribution $\pi = \left(\frac{1}{2}, \frac{1}{2} \right)$

and transition matrix $P = \begin{pmatrix} 0.1 & 0.9 \\ 0.3 & 0.7 \end{pmatrix}$

(a.) Chain is irreducible.

(b.) Chain is aperiodic.

(c.) State 1 is transient.

(d.) State 0 is recurrent.

5. Let $\{X_n; n \geq 1\}$ be a Markov chain with state space $S = \{0, 1, 2, 3, 4\}$ and transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 2/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Which of the following is true?

(a.) $C(0) = \{0, 3\}$.

(b.) State 2, 3 are aperiodic.

(c.) Chain is irreducible.

(d.) State 1 is recurrent.

6. Consider the Markov chain with state space $S = \{0, 1, 2, 3\}$ and transition matrix

$$P = \begin{pmatrix} 0.8 & 0 & 0.2 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0 & 0.3 \end{pmatrix}$$

Which of the following is true?

(a.) State 0 is recurrent.

(b.) State 2 is recurrent.

(c.) State 1 is transient.

(d.) State 4 is recurrent.

7. Consider a Markov Chain with state space $S = \{0, 1, 2, 3\}$ and with transition probability matrix P is given by

$$P = \begin{pmatrix} 2/3 & 0 & 1/3 & 0 \\ 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}$$

Then

(a.) 1 is a recurrent state.

(b.) 0 is a recurrent state.

(c.) 3 is a recurrent state.

(d.) 2 is a recurrent state.

8. Consider a Markov Chain with state space $\{1, 2, \dots, 100\}$. Suppose states $2i$ and $2j$ communicate with each other and states $2i-1$ and $2j-1$ communicate with each other for every $i, j = 1, 2, \dots, 50$. Further, suppose that $P_{3,3}^{(2)} > 0, P_{4,4}^{(3)} > 0$ and $P_{2,5}^{(7)} > 0$. Then

- (a.) The Markov Chain is irreducible.
(b.) The Markov Chain is a periodic.
(c.) State 8 is recurrent.
(d.) State 9 is recurrent.

9. Consider a Markov Chain on the state space $\{1, 2, 3, 4, 5\}$ with transition probability matrix

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 & 1/4 & 0 \\ 1/3 & 0 & 0 & 2/3 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 2/5 & 1/5 & 0 & 2/5 & 0 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \end{pmatrix}$$

Then which of the following statements are true?

- (a.) States 1, 2, 4 are recurrent and 3, 5 are transient.
(b.) States 1, 2, 3, 4 are recurrent and state 5 is transient.
(c.) The chain has a unique stationary distribution.
(d.) The chain has more than one stationary distributions.

10. Let $\{P_n; n \geq 0\}$ be a sequence of numbers such that for all $n \geq 0$, $P_n > 0$, $\sum_{n=0}^{\infty} P_n = 1$,

$\sum_{n=0}^{\infty} n P_n < \infty$. Consider a Markov chain on state space $S = \{0, 1, 2, \dots\}$ with transition probability matrix

$$P = \begin{pmatrix} P_0 & P_1 & P_2 & \dots \\ 1 & 0 & 0 & \dots \\ 0 & 1 & 0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

- (a.) The chain is not irreducible.
(b.) The chain is irreducible and transient.
(c.) The chain is irreducible and null recurrent.
(d.) The chain is irreducible and positive recurrent.

11. One choose letters with replacement from the set $\{a, b, c, d\}$ as follows. After one letter is observed at step n , in $(n+1)^{\text{th}}$ step another letter will be chosen from the other three with equal probability $\frac{1}{3}$. Let X_n denote the letter

chosen at the n^{th} step. Which of the following is necessarily true for the Markov Chain X_n ?

(a.) $\lim_{n \rightarrow \infty} P(X_n = a) = \frac{1}{3}$

(b.) $\lim_{n \rightarrow \infty} P(X_n = b) = \frac{1}{4}$

(c.) the average proportion of time 'c' is observed converges to $\frac{1}{4}$

(d.) chain is irreducible

12. consider a Markov chain $\{X_n : n \in \mathbb{N}\}$ such that $P_{ii} > 0$ for some i . Let chain is irreducible then

- (a.) chain is aperiodic always
(b.) chain can never be aperiodic
(c.) chain may be aperiodic or may be periodic
(d.) chain has period 2.

13. The one step transition probability matrix of a homogeneous Markov chain $\{X_n, n \geq 0\}$ is

$$\begin{bmatrix} 1/3 & 1/6 & 1/6 & 1/3 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/5 & 1/4 & 1/4 & 3/10 \\ 1/6 & 1/6 & 1/3 & 1/3 \end{bmatrix}$$

Then which of the following is correct?

- (a.) $P(X_7 = 1 | X_5 = 2) = P(X_{11} = 1 | X_9 = 2)$
(b.) this Markov chain is irreducible
(c.) the states 1 and 2 are the only recurrent states
(d.) it is a recurrent Markov chain

14. The one step transition probability matrix of a homogeneous Markov chain $\{X_n, n \geq 0\}$ is

$$\begin{bmatrix} 1/3 & 1/6 & 1/6 & 1/3 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/5 & 1/4 & 1/4 & 3/10 \\ 1/6 & 1/6 & 1/3 & 1/3 \end{bmatrix}$$

Then which of the following is correct?

- (a.) $P(X_7 = 1 | X_5 = 2) = P(X_{11} = 1 | X_9 = 2)$
(b.) this Markov chain is irreducible
(c.) the states 1 and 2 are the only recurrent states
(d.) it is a recurrent Markov chain

15. Consider an aperiodic Markov chain with state space S and with stationary transition probability matrix $P = (p_{ij})$, $ij \in S$. Let the n -step transition probability matrix be denoted by $P^n = (p_{ij}^n)$, $i, j \in S$. Then which of the following statements is true?

- (a.) $\lim_{n \rightarrow \infty} p_{ii}^n = 0$ Only if i is transient
(b.) $\lim_{n \rightarrow \infty} p_{ii}^n > 0$ and only if i is recurrent
(c.) $\lim_{n \rightarrow \infty} p_{ij}^n = \lim_{n \rightarrow \infty} p_{jj}^n$ if i and j are in the same communicating class.
(d.) $\lim_{n \rightarrow \infty} p_{ij}^n = \lim_{n \rightarrow \infty} p_{ii}^n$ if i and j are in the same communicating class.

16. Let $\{X_n\}$ be a stationary Markov chain such that

$$P(X_{i+1} = 1 | X_i = 1) = p_1 = 1 - P(X_{i+1} = 0 | X_i = 1)$$

$$P(X_{i+1} = 1 | X_i = 0) = p_0 = 1 - P(X_{i+1} = 0 | X_i = 0),$$

and $P(X_1 = 1) = \pi_1 = 1 - P(X_1 = 0)$

Then

- (a.) $\pi_1 = p_1$
(b.) $\pi_1 = p_0$
(c.) $\pi_1 = \frac{p_0}{1 - p_1 + p_0}$
(d.) $\pi_1 = \frac{1}{2}$

17. An aperiodic Markov chain with stationary transition probabilities on the state space $\{1, 2, 3, 4, 5\}$ must have

- (a.) At least one null recurrent state
(b.) At least one positive recurrent state
(c.) At least one positive recurrent and at least one null recurrent state
(d.) At least one transient state

18. Let P be the stationary transition probability matrix of the Markov Chain $\{X_n, n \geq 0\}$, which is irreducible and every state has period 2. Further suppose that the Markov chain $\{Y_n, n \geq 0\}$ on the same state space has transition probability matrix P^2 . Both the chains are assumed to have the same initial distribution. Then

- (a.) $P[X_0 = Y_0, X_2 = Y_2] = 1$
(b.) All states of the chain $\{Y_n, n \geq 0\}$ are aperiodic
(c.) The chain $\{Y_n, n \geq 0\}$ is irreducible
(d.) If a state is recurrent for the chain $\{X_n, n \geq 0\}$, then it is also recurrent for the chain $\{Y_n, n \geq 0\}$

19. Let $\mu = (\mu_1, \mu_2, \mu_3, \mu_4)$ be the stationary distribution for a Markov chain on the state space $\{1, 2, 3, 4\}$ with transition probability matrix P . Suppose that the states 1 and 2 are transient and the states 3 and 4 form a communicating class. Which of the following is/are true?

- (a.) $\mu P^3 = \mu P^5$
(b.) $\mu_1 = 0$ and $\mu_2 = 0$
(c.) $\mu_3 + \mu_4 = 1$
(d.) One of μ_3 and μ_4 is zero.

20. Consider a Markov chain with state space $S = \{1, 2, 3, 4, 5\}$ and stationary transition probability matrix P given by

$$P = \begin{pmatrix} 0.1 & 0 & 0.2 & 0.7 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0.7 & 0 & 0.1 & 0.2 & 0 \\ 0.2 & 0 & 0.7 & 0.1 & 0 \\ 0 & 0.5 & 0 & 0 & 0.5 \end{pmatrix}. \text{ Let } p_n^{(n)} \text{ be the}$$

(i, j) th element of P^n . Then

- (a.) $\sum_{i=1}^5 \lim_{n \rightarrow \infty} p_n^{(n)} = 1$
(b.) $(0.25, 0.25, 0.25, 0.25, 0)$ is a stationary distribution for the Markov chain.
(c.) $\sum_{n=1}^{\infty} p_{33}^{(n)} < \infty$
(d.) $\lim_{n \rightarrow \infty} p_{14}^{(n)} = \frac{1}{3}$

21. Let $\{p_n, n \geq 0\}$ be a sequence of numbers, such that for all $n \geq 0$, $p_n > 0$, $\sum_{n=0}^{\infty} p_n = 1$ and

$\sum_{n=0}^{\infty} np_n < \infty$. Consider a Markov chain on the state space $\{0, 1, 2, \dots\}$ with transition

probability matrix $\begin{pmatrix} p_0 & p_1 & p_2 & \dots \\ 1 & 0 & 0 & \dots \\ 0 & 1 & 0 & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$. Then

- (a.) The chain is not irreducible
(b.) The chain is irreducible and transient
(c.) The chain is irreducible and null recurrent
(d.) The chain is irreducible and positive recurrent.

22. Consider a Markov chain $\{X_n : n \geq 0\}$ on the state space $\{0, 1\}$ with transition probability matrix P . Which of the following statements are necessarily true?
- (a.) When $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\lim_{n \rightarrow \infty} P_v[X_n = i]$ converges for $i = 0, 1$, but the limits depend on the initial distribution v .
- (b.) When $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\lim_{n \rightarrow \infty} P_v[X_n = 1]$ exists and is positive for all choices for the initial distribution v .
- (c.) When $P = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 1 \end{pmatrix}$, $\lim_{n \rightarrow \infty} P_v[X_n = 0]$ does not exist for any choice of the initial distribution v .
- (d.) When $P = \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix}$, $\lim_{n \rightarrow \infty} P_v[X_n = 0]$ always exists, but may be 0 for some choice of the initial distribution v .
23. Let X_n be the result of the n -th roll of a fair die, $n \geq 1$.
Let $S_n = \sum_{i=1}^n X_i$ and Y_n be the last digit of S_n , for $n \geq 1$ and $Y_0 = 0$.
Then, which of the following statements are correct?
- (a.) $\{Y_n : n \geq 0\}$ is an irreducible Markov chain
- (b.) $\{Y_n : n \geq 0\}$ is an aperiodic Markov chain
- (c.) $P(Y_n = 0) \rightarrow \frac{1}{6}$ as $n \rightarrow \infty$
- (d.) $P(Y_n = 0) \rightarrow \frac{1}{10}$ as $n \rightarrow \infty$
24. Let $\{X_n : n \geq 0\}$ be a Markov chain on a finite state space S with stationary transition probability matrix. Suppose that the chain is not irreducible. Then the Markov chain
- (a.) Admits infinitely many stationary distributions
- (b.) Admits a unique stationary distribution
- (c.) May not admit any stationary distribution
- (d.) Cannot admit exactly two stationary distributions

ASSIGNMENT SHEET-2

1. Suppose that the random variable X has a uniform distribution, $X \sim U[0, 1]$, and that, once $X = x$ has been observed, the conditional distribution of Y is $[Y|X=x] \sim U[x, 1]$. Then which of the following is/are true?

(a) $E(Y) = \left(\frac{x+1}{2}\right)$

(b) $E(Y) = \left(\frac{3}{4}\right)$

(c) $V(Y) = \left(\frac{1}{36}\right)$

(d) $V(Y) = \left(\frac{7}{144}\right)$

2. Suppose that the random variable X and Y have a continuous joint distribution with pdf $f(x, y)$, means μ_X and μ_Y respectively, variances σ_X^2 & σ_Y^2 respectively and correlation ρ .

Suppose also that

$$E(Y|x) = \beta_0 + \beta_1 x$$

Which of the following is/are true?

(a) $\int_{-\infty}^{\infty} y f(x, y) dy = (\beta_0 + \beta_1 x) f_X(x)$

(b) $\mu_Y = \beta_0 + \beta_1 \mu_X$

(c) $\sigma_X \sigma_Y + \mu_X \mu_Y = \beta_0 \mu_X + \beta_1 (\sigma_X^2 + \mu_X^2)$

(d) $\beta_1 = \frac{\rho \sigma_Y}{\sigma_X}$ and $\beta_0 = \mu_Y - \beta_1 \mu_X$

3. Suppose that a random sample is to be taken from a normal distribution for which the value of the mean θ is unknown and the standard deviation is 2. How large a random sample must be taken in order that

$$P(|\bar{X} - \theta| < 0.1) \geq 0.95 \quad \forall \theta$$

Note that: $Z_{0.025} = 1.96$ i.e.,

$$P(Z \geq 1.96) = 0.025$$

- (a) 40
(b) 25
(c) 50
(d) 30

4. let Y_1 and Y_2 be two random variables with joint density

$Y_1 \backslash Y_2$	0	1	2
0	q^3	pq^2	0
1	pq^2	pq	p^2q
2	0	p^2q	p^3

then

(a) $P(Y_1 = 2) = p^2$

(b) $P(Y_2 = 0) = q^2$

(c) $E(Y_1 - Y_2) = 0$

(d) Y_1 and Y_2 are Binomial random variables

5. Suppose that X is a random variable having finite mean μ and variance σ^2 . Then:

(a) $P(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$ for $\epsilon > 0$

(b) $P(X - \mu \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$ for $\epsilon > 0$

(c) $P(|X - \mu| \geq \epsilon) \leq 1 - \frac{\sigma^2}{\epsilon^2}$ for $\epsilon > 0$

(d) $P(X - \mu \geq \epsilon) \leq 1 - \frac{\sigma^2}{\epsilon^2}$ for $\epsilon > 0$

6. Let $\{X_n\}$ be any sequence of random variables. Then for the sequence $\{X_n\}$ to satisfy the weak law of large numbers, the condition, for $Y_n = n^{-1} \sum_{k=1}^n X_k$ that

$$E\left\{\frac{Y_n^2}{1 + Y_n^2}\right\} \rightarrow 0 \text{ as } n \rightarrow \infty \text{ is:}$$

- (a) necessary and sufficient condition
(b) necessary condition
(c) sufficient condition
(d) neither necessary nor sufficient condition

7. If X is a gamma variate with parameters 1 and β then:

- (a) the random variable X has no memory
(b) $P(X > r+s | X > s) = P(X > r)$
(c) X follows exponential distribution with parameter β
(d) all the above are true

8. If X and Y are two standard normal variates and ρ is the correlation coefficient between X and Y , then the correlation coefficient between X^2 and Y^2 is:
- ρ
 - ρ^2
 - $\rho(1-\rho^2)$
 - none of the above is true.
9. Let X and Y be two variables which can be correlated by an equation $aX + bY + c = 0$. If a and b are of the different signs then the correlation coefficient between X and Y is:
- 1
 - 0
 - 1
 - Between -1 and 1.
10. A system of 5 identical units consists of two parts A and B which are connected in series. Part A has 2 units connected in parallel and part B has 3 units connected in parallel. All the 5 units function independently with probability of failure $\frac{1}{2}$. Then the reliability, of the system is
- $\frac{31}{32}$
 - $\frac{11}{32}$
 - $\frac{1}{32}$
 - $\frac{21}{32}$
11. Suppose $X_1, X_2, \dots$ is an i.i.d. sequence of random variables with common variance $\sigma^2 > 0$. Let $Y_n = \frac{1}{n} \sum_{i=1}^n X_{2i-1}$ and $Z_n = \frac{1}{n} \sum_{i=1}^n X_{2i}$. Then the asymptotic distribution (as $n \rightarrow \infty$) of $\sqrt{n}(Y_n - Z_n)$ is
- $N(0, 1)$
 - $N(0, \sigma^2)$
 - $N(0, 2\sigma^2)$
 - Degenerate at 0
12. Suppose X is a random variable with $E(X) = \text{Var}(X)$. Then the distribution of X
- Is necessarily Poisson
 - Is necessarily Exponential
 - Is necessarily Normal
 - Cannot be identified from the given data.
13. Batteries for torch lights are packed in boxes of 10 and a lot contains 10 boxes. A quality inspector randomly chooses a box and then checks two batteries selected randomly without replacement from that box. The lot will be rejected if any one of the two chosen batteries turns out to be defective. Suppose that 9 of the 10 boxes in the lot contain no defective batteries and only one box contains 2 defective ones. What is the probability that the lot will NOT be passed by the Inspector?
- $\frac{197}{4950}$
 - $\frac{98}{2475}$
 - $\frac{8}{225}$
 - $\frac{17}{450}$
14. Which of the following is/are cumulative distribution function(s) (c.d.f.) of random variable(s)?
- $F_1(x) = \begin{cases} 0, & x \leq 0 \\ e^{-x}, & x > 0 \end{cases}$
 - $F_2(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x}, & x > 0 \end{cases}$
 - $F_3(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$
 - $F_4(x) = \begin{cases} 0, & x < 0 \\ 1/2, & 0 \leq x < 1 \\ 1, & x \geq 0 \end{cases}$
15. Let X be a random variable taking values in a set E . Let $P(x > a + b | X > a) = P(X > b)$ for all $a, b \in E$. Then which of the following is a possible distribution of X ?
- Poisson
 - Geometric
 - Long-normal
 - Exponential

16. Suppose X and Y are independent $N(0,1)$ random variables.

Let $U = \frac{X}{Y}$ and $V = \frac{X}{|Y|}$. Then

- (a.) U and V are independent
(b.) U and V have the same distribution
(c.) $P(U=V) = \frac{1}{2}$
(d.) $P(U < V) = \frac{1}{2}$

17. Suppose $X_1, X_2, \dots$ is a sequence of i.i.d. random variables where

$$P(X_i = 1) = p = 1 - P(X_i = 0), \quad i = 1, 2, \dots$$

Let $Z = \frac{1}{500} \sum_{i=1}^{500} X_i$ and $\alpha = P(|Z - p| > 0.1)$

Then for all p

- (a.) $\alpha \leq .1$
(b.) $\alpha \leq .05$
(c.) $\alpha > .01$
(d.) $\alpha = 0$

18. Random variables X_1, X_2, X_3 are such that correlation $(X_1, X_2) = \text{correlation}(X_2, X_3) = \text{correlation}(X_3, X_1) = \rho$

- (a.) ρ cannot be negative
(b.) ρ can take any value between -1 and $+1$
(c.) $\rho \geq -0.5$
(d.) ρ is either $+1$ or -1

19. Suppose that we have a data set consisting of 25 observations, where each value is either 5 or 10.

- (a.) The mean of the data cannot be larger than the median.
(b.) The mean of the data cannot be smaller than the median.
(c.) The mean and the median for the data will be the same only if the variance of the data is zero
(d.) The mean and the median for the data will be different only if the range is 5.

20. Suppose $X \sim N(0, 1)$ and $Y \sim \chi_n^2$. Which of the following is always correct?

- (a.) $X^2 + Y \sim \chi_{n+1}^2$
(b.) $\frac{X}{\sqrt{\frac{Y}{n}}} \sim t_n$
(c.) $E(X^2 + Y) = 1 + n$
(d.) $\text{Var}(X + Y) = 1 + 2n$

21. Let $X_1, X_2, \dots$ be i.i.d. random variables with mean 0 and variance 2 and let $Y_i = \frac{X_i + X_{i+1}}{2}$ for $i \geq 1$. Then the limiting distribution of

- (a.) $\frac{1}{\sqrt{2n}}(Y_1 + Y_2 + \dots + Y_n)$ is normal with mean 0 and variance 1
(b.) $\frac{1}{\sqrt{n}}(Y_1 + Y_3 + Y_5 + \dots + Y_{2n-1})$ is normal with mean 0 and variance 1
(c.) $\frac{1}{\sqrt{n}}(Y_2 + Y_4 + Y_6 + \dots + Y_{2n})$ is normal with mean 0 and variance 1
(d.) $\frac{1}{\sqrt{n}}(Y_1 + Y_2 + Y_3 + \dots + Y_{2n-1})$ is normal with mean 0 and variance 1

22. $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ are independent and identically distributed random vectors where X_i is normally distributed with mean 0 and variance 1 and

$P[Y_i = 2] = P[Y_i = -2] = \frac{1}{2}$. Further, x_i and Y_i are independently distributed for

$i = 1, \dots, n$ and

$$Z_n = X_1 Y_1 - X_2 Y_2 + \dots + (-1)^{n-1} X_n Y_n. \text{ Then}$$

- (a.) Z_n is normally distributed for each n
(b.) Z_n is symmetrically distributed about 0
(c.) $V\left(\frac{Z_n}{\sqrt{n}}\right) = 4$
(d.) $V\left(\frac{Z_n}{\sqrt{n}}\right) = 2$

23. Let $F(x, y)$, $G(x)$ and $H(y)$ be the joint c. d. f. of (X, Y) , marginal c. d. f. of X and marginal c. d. f. of Y respectively. Define $U = \begin{cases} 1 & \text{if } X \leq a \\ -1 & \text{if } X > a \end{cases}$ and $V = \begin{cases} 1 & \text{if } Y \leq b \\ -1 & \text{if } Y > b \end{cases}$. Where a and b are fixed real numbers. Then
- If $\text{Cov}(U, V) = 0$ then $F(x, y) = G(x)H(y)$ for all x and y .
 - If $F(x, y) = G(x)H(y)$ for all x and y then $\text{Cov}(U, V) = 0$
 - If U and V are independent then X and Y are independent.
 - If X and Y are independent then U and V are independent.
24. Suppose X and Y are independent random variables where Y is symmetric about 0. Let $U = X + Y$ and $V = X - Y$. Then
- U and V are always independent
 - U and V have the same distribution
 - U is always symmetric about 0
 - V is always symmetric about 0
25. Let $X_1, X_2, \dots$ be i.i.d. $N(1, 1)$ random variables. Let $S_n = X_1^2 + X_2^2 + \dots + X_n^2$ for $n \geq 1$. Then $\lim_{n \rightarrow \infty} \frac{\text{Var}(S_n)}{n}$ is
- 4
 - 6
 - 1
 - 0
26. Which of the following conditions imply independence of the random variables X and Y ?
- $p(X > a | Y > a) = P(X > a)$ For all $a \in \mathbb{R}$.
 - $p(X > a | Y < b) = P(X > a)$ for all $a, b \in \mathbb{R}$
 - X and Y are uncorrelated.
 - $E[(X - a)(Y - b)] = E(X - a)E(Y - b)$ for all $a, b \in \mathbb{R}$
27. Suppose that we have a data set consisting of 25 observations, where each value is either 0 or 1.
- The mean of the data cannot be larger than the variance.
 - The mean of the data cannot be smaller than the variance.
 - The mean being same as the variance implies that the mean is zero.
 - The variance will be 0 if and only if the mean is either 1 or 0.
28. Consider the following five observations on (X, Y) ; $(0, 1), (1, 2), (2, 3), (3, 2), (4, 1)$. Then
- The least-square linear regression of Y on X is $Y = \frac{9}{5}$
 - The least-square linear regression of X on Y is $X = 2$
 - The correlation coefficient between X and Y is 0.
 - The correlation coefficient between X and Y is +1
29. Let $X_1, X_2, \dots$ be independent random variables with X_n being uniformly distributed between $-n$ and $3n, n = 1, 2, \dots$. Let $S_n = \frac{1}{\sqrt{N}} \sum_{n=1}^N \frac{X_n}{n}$ for $N = 1, 2, \dots$ and let F_N be the distribution function of S_N also let ϕ denote the distribution function of a standard normal random variable. Which of the following is/are true?
- $\lim_{N \rightarrow \infty} F_N(0) \leq \phi(0)$
 - $\lim_{N \rightarrow \infty} F_N(0) \geq \phi(0)$
 - $\lim_{N \rightarrow \infty} F_N(1) \leq \phi(1)$
 - $\lim_{N \rightarrow \infty} F_N(1) \geq \phi(1)$
30. Let $X_1, X_2, \dots$ be i.i.d. standard normal random variables and let $T_n = \frac{X_1^2 + \dots + X_n^2}{n}$. Then
- The limiting distribution of $T_n - 1$ is χ^2 with 1 degree of freedom.
 - The limiting distribution of $\frac{T_n - 1}{\sqrt{n}}$ is normal with mean 0 and variance 2.
 - The limiting distribution of $\sqrt{n}(T_n - 1)$ is χ^2 with 1 degree of freedom.
 - The limiting distribution of $\sqrt{n}(T_n - 1)$ is normal with mean 0 and variance 2.
31. Suppose X_1, X_2, X_3, X_4 are i.i.d. random variables taking values 1 and -1 with probability $\frac{1}{2}$ each. Then $E(X_1 + X_2 + X_3 + X_4)^4$ equals
- 40
 - 76
 - 16
 - 12

32. Suppose that $\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ has Normal $(\mu_{2 \times 1}, \Sigma_{2 \times 2})$ distribution, where $\Sigma_{2 \times 2}$ is non-singular. Let $X_3 = -2X_2$. Then which of the following has a singular normal distribution?
- $\begin{pmatrix} X_1 - 2X_2 \\ X_2 - 2X_3 \end{pmatrix}$
 - $\begin{pmatrix} X_1 - X_2 - X_3 \\ 2X_1 + 2X_2 \end{pmatrix}$
 - $\begin{pmatrix} X_1 + X_2 \\ 2X_1 + 2X_3 \end{pmatrix}$
 - $\begin{pmatrix} X_1 + X_2 + X_3 \\ X_1 + X_2 \end{pmatrix}$
33. Let X be a binomial random variable with parameters $\left(11, \frac{1}{3}\right)$. At which value(s) of k is $P(X = k)$ maximized?
- $k = 2$
 - $k = 3$
 - $k = 4$
 - $k = 5$
34. X, Y, Z are independent random variables with $N(0, 1)$ (standard normal) distribution. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 1$, if $x \geq 0$ and $f(x) = -1$, if $x < 0$. Let U, V, W be defined by $U = |X| \cdot f(Y)$, $V = |Y| \cdot f(X)$, $W = Z \cdot f(X)$. Then
- U and V are independent each having a $N(0, 1)$ distribution.
 - U and W are independent each having $N(0, 1)$ distribution.
 - V and W are independent each having $N(0, 1)$ distribution.
 - U, V and W are independent random variables.
35. Let X_1 and X_2 be two independent random variables with $X_1 \sim \text{binomial}\left(m, \frac{1}{2}\right)$ and $X_2 \sim \text{binomial}\left(n, \frac{1}{2}\right)$, $m \neq n$. Which of the following are always true?
- $2X_1 + 3X_2 \sim \text{binomial}\left(2m + 3n, \frac{1}{2}\right)$
 - $X_2 - X_1 + m \sim \text{binomial}\left(m + n, \frac{1}{2}\right)$
 - Conditional distribution of X_2 given $(X_1 + X_2)$ is hyper-geometric
 - Distribution of $X_1 - X_2$ is symmetric about 0.
36. Let A be an event, which has non zero probability such that $P(A \cap B_i) = P(B_i)P(A|B_i)$ and $P(A \cap B_i) = P(A)P(B_i|A)$. Then $B_1, B_2, \dots, B_n$ are:
- exhaustive, may not be mutually exclusive and none of which has zero probability.
 - exhaustive, may not be mutually exclusive and atleast one of which has a non-zero probability.
 - exhaustive, mutually exclusive and atleast one of which has a non-zero probability.
 - exhaustive, mutually exclusive and none of which has zero probability.
37. Let n balls be put at random into n cells, then the probability that each cell will be occupied is:
- $1/n^n$
 - $1/n$
 - $n!/n^n$
 - $1/n!$
38. Let $X_1, X_2, \dots, X_n$ be n independent observations from distribution with density function $f(\cdot)$ and distribution function $F(\cdot)$. Then the density function $Y = \min(X_1, X_2, \dots, X_n)$
- $n[F(y)]^{n-1} f(y)$
 - $n[1 - F(y)]^{n-1} f(y)$
 - $[1 - F(y)]^n f(y)$
 - $[F(y)]^n f(y)$
39. If $p(x) = 2^{-x}$, $x = 1, 2, 3, \dots$, then $P[|X - 2| \leq 2]$:
- cannot be computed from Chebychev's inequality.
 - can only be computed from Chebychev's inequality.
 - computed from Chebychev's inequality gives a result identical to that obtained from direct computation.
 - computed from Chebychev's inequality gives as precise a result as obtained from direct computation.

40. If X and Y are two independent non-negative integer-valued random variables such that,
 $P(X=k) > 0$ and $P(Y=k) > 0$ for $k = 0, 1, 2, \dots$
 and the conditional distribution of X given $X+Y$ is binomial, then:
 (a.) both X and Y are binomial.
 (b.) X is binomial and Y is Poisson.
 (c.) X is Poisson and Y is binomial.
 (d.) both X and Y are Poisson.
41. If X follows a binomial distribution with parameters n and p , then $Z = \frac{(X-np)}{npq}$ is asymptotically normal if:
 (a.) n is large and neither p nor q is close to one.
 (b.) n is large and neither p nor q is close to zero.
 (c.) n is small and neither p nor q is close to zero.
 (d.) n is small and neither p nor q is close to one.
42. Let X be a Poisson random variable with parameter λ . Then as $\lambda \rightarrow \infty$:
 (a.) $(X-\lambda)/\lambda$ is asymptotically normal.
 (b.) $(X-\sqrt{\lambda})/\sqrt{\lambda}$ is asymptotically normal.
 (c.) $(X-\lambda)/\sqrt{\lambda}$ is asymptotically normal.
 (d.) none of the above.
43. Let there be N units in a population and let M units be added to it before a sample of size n is selected. Then the probability of the selection of any unit at a specified draw in simple random sampling is:
 (a.) $\frac{n}{N+M}$
 (b.) $\frac{1}{N+M}$
 (c.) $\frac{N+M}{C_n}$
 (d.) $\frac{1}{N+M C_n}$
44. Let $X_i, (i=1, 2, \dots, n)$ be a random sample of size n drawn from a normal population having mean μ and variance σ^2 , then for $s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$, the distribution of the statistic $\frac{\sqrt{n}(\bar{X} - \mu)}{s}$ is:
 (a.) student's t with $(n-1)$ degrees of freedom.
 (b.) normal.
 (c.) chi-square.
 (d.) standard normal.
45. Let t be any statistics, $E(t)$ be the hypothetical or expected value of t , $V(t)$ be the variance of t and $SE(t)$ be the standard error of t , then Z is normal with mean 0 and variance 1, where Z is:
 (a.) $[t - E(t)]/V(t)$
 (b.) $[t - E(t)]/\sqrt{SE(t)}$
 (c.) $[t - E(t)]/SE(t)$
 (d.) $[t - E(t)]/SE(t)$
46. $X \sim U(0, 1)$ and $Y \sim B(10, X)$, then $V(Y)$ is
 (a.) 5
 (b.) 6
 (c.) 10
 (d.) 12
47. Which one of the following statements is false?
 Characteristic functions are:
 (a.) independent of origin and scale.
 (b.) independent of origin and not of scale.
 (c.) independent of scale and not of origin.
 (d.) independent neither of origin nor of scale.

48. Which one of the following statements is true?

(a.) Correlation coefficient between X and Y is independent of units of measurements of the variables X and Y .

(b.) Let $\rho(X, Y)$ be the correlation coefficient between X and Y . Then the correlation between U and V is the same as that between X and Y , where:

$$U = \frac{(X-a)}{c} \text{ and } V = \frac{(Y-b)}{d}$$

for a, b, c and d to be four arbitrarily chosen constants.

(c.) Correlation coefficient is independent of origin and scale.

(d.) If $x'_i = \frac{(x_i - \bar{x})}{\sigma_1}$ and $y'_i = \frac{(y_i - \bar{y})}{\sigma_2}$, where

σ_1 and σ_2 are standard deviations of X

and Y respectively, then $\frac{1}{n} \sum_{i=1}^n x'_i y'_i$,

which lie between -1 and 1 , is also a correlation coefficient.

49. Which one of the following statements is true?

(a.) If X and Y are two independent random variables and $Z = X + Y$ is chi-square distributed with n degrees of freedom. For X to be chi-square distributed with n_1 degrees of freedom, Y should also be chi-square distributed with $n - n_1$ degrees of freedom for $n > n_1$.

(b.) For large values of n , $\sqrt{2\chi^2} - \sqrt{2n-1}$ is nearly normally distributed with mean 0 and variance 1.

(c.) The mean and variance of a chi-square distribution with n degrees of freedom are n and $2n$ respectively.

(d.) For a chi-square distribution with n degrees of freedom, the moment generating and the characteristic functions are $(1-2t)^{n/2}$ and $(1-2it)^{-n/2}$ respectively.

50. Let $Y_1 < Y_2 < \dots < Y_n$ be the order statistics from uniform distribution over $(0,1)$, then the correlation coefficient between Y_1 and Y_n is

(a.) 1

(b.) 0

(c.) directly proportional to sample size

(d.) inversely proportional to sample size

51. If $X \sim B(n, p)$, then $\text{Cov}\left(\frac{X}{n}, \frac{n-X}{n}\right)$ is

(a.) $-\frac{pq}{n}$

(b.) $\frac{pq}{n}$

(c.) $-\frac{p}{n}$

(d.) $-\frac{q}{n}$

52. Let X be a positive integer valued random variable. Then its expected value is given by

(a.) $\sum_{k=1}^{\infty} P[X > k]$

(b.) $\sum_{k=1}^{\infty} P[X \geq k]$

(c.) $\sum_{k=1}^{\infty} P[X \leq k]$

(d.) $\sum_{k=1}^{\infty} P[X = k]$

53. Let X_1 and X_2 be two independent standard normal variables. Let $Y = \frac{(X_2 - X_1)^2}{2}$. Then $V(Y)$ is

(a.) 1

(b.) 2

(c.) $1/2$

(d.) None of these

54. Hundred (100) tickets are marked $1, 2, \dots, 100$ and are arranged at random. Four tickets are picked from these tickets and are given to four persons A, B, C and D . What is the probability that A gets the ticket with the largest value (among A, B, C, D) and gets the ticket with the smallest value (among A, B, C, D)?

(a.) $\frac{1}{4}$

(b.) $\frac{1}{6}$

(c.) $\frac{1}{2}$

(d.) $\frac{1}{12}$

55. Let X and Y be independent and identically distributed random variables such that

$$P(X=0)=P(X=1)=\frac{1}{2}. \text{ Let } Z=X+Y \text{ and}$$

$W=|X-Y|$. Then which statement is not correct?

- (a.) X and W are independent.
- (b.) Y and W are independent.
- (c.) Z and W are independent.
- (d.) Z and W are independent.

56. Let $X_1 \sim N(0,1)$ and let

$$X_2 = \begin{cases} -X_1, & -2 \leq X_1 \leq 2 \\ X_1, & \text{otherwise} \end{cases}. \text{ Then identify the correct statement.}$$

- (a.) $\text{corr}(X_1, X_2) = 1$.
- (b.) X_2 does not have $N(0,1)$ distribution
- (c.) (X_1, X_2) has a bivariate normal distribution.
- (d.) (X_1, X_2) does not have a bivariate normal distribution.

57. Let $\underline{X} \sim N_3(\underline{\mu}, \underline{\Sigma})$ where $\underline{\mu} = (1, 1, 1)$ and

$$\underline{\Sigma} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & c \\ 1 & c & 2 \end{pmatrix}. \text{ The value of } c \text{ such that } X_2$$

and $-X_1 + X_2 - X_3$ are independent is

- (a.) -2
- (b.) 0
- (c.) 2
- (d.) 1

58. Let $(\Omega, \mathcal{F}, P)$ be a probability space and let A be an event with $P(A) > 0$. In which of the following cases does Q define a probability measure on $(\Omega, \mathcal{F})$?

- (a.) $Q(D) = P(A \cup D) \forall D \in \mathcal{F}$
- (b.) $Q(D) = P(A \cap D) \forall D \in \mathcal{F}$
- (c.) $Q(D) = \begin{cases} P(A|D), & \text{if } D \in \mathcal{F} \text{ with } P(D) > 0 \\ 0, & \text{if } P(D) = 0 \end{cases}$
- (d.) $Q(D) = P(D|A) \forall D \in \mathcal{F}$

59. The joint probability density function of (X, Y) is

$$f(x, y) = \begin{cases} 6(1-x), & 0 < y < x, 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

- (a.) X and Y are not independent.

$$(b.) f_y(y) = \begin{cases} 3(y-1)^2, & 0 < y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

- (c.) X and Y are independent.

$$(d.) f_y(y) = \begin{cases} 3(y - \frac{1}{2}y^2), & 0 < y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

60. Suppose X and Y are independent and identically distributed random variables and let $Z = X + Y$. Then the distribution of Z is in the same family as that of X and Y if X is

- (a.) Normal
- (b.) Exponential
- (c.) Uniform
- (d.) Binomial

ASSIGNMENT SHEET-3

1. If $X_1, X_2, \dots, X_n$ are n uncorrelated random variables following the same distribution, then the correlation coefficient between $\bar{X}$ and $X_i - \bar{X}$ is:
 - (a.) 0
 - (b.) 1
 - (c.) $1/2$
 - (d.) Non-existent.
2. Both type I and type II errors are bad and these should be reduced to the minimum. Nevertheless, which one of the following statements about these errors is false?
 - (a.) These errors have no relevance in census.
 - (b.) For a given sample size, these two errors neutralise each other, ... precisely, if probability of occurrence of type I error is reduced, the probability of type II error would simultaneously be increased.
 - (c.) During testing procedures, type I error being assumed to be more severe, is controlled.
 - (d.) All the above statements are false.
3. $X_1, \dots, X_n$ is a random sample from the $N(\mu, \sigma^2)$ population. If $T_1 = X_1 + \dots + X_n$ and $T_2 = X_1^2 + \dots + X_n^2$ then which of the following statements is correct regarding sufficient statistics for μ and σ^2
 - (a.) T_1 is sufficient statistic for μ
 - (b.) T_2 is sufficient statistic for σ^2
 - (c.) $T_2 - T_1$ is sufficient for (μ, σ^2)
 - (d.) (T_1, T_2) is sufficient for (μ, σ^2)
4. A random sample of size ' n ' is drawn from a distribution with density function $f(x; \theta) = \theta(1-x)^{1-\theta}$; $0 < x < 1$; $0 < \theta < \infty$. A sufficient statistic for θ is
 - (a.) Sample mean
 - (b.) Sample median
 - (c.) $\sum_{i=1}^n \log(1-x_i)$
 - (d.) $\sum_{i=1}^n x_i^2$
5. X_1, X_2 are two independent observations from $B(1, \theta)$. The statistic $X_1 - X_2$ is
 - (a.) Sufficient and complete
 - (b.) Neither sufficient nor complete
 - (c.) Complete but not sufficient
 - (d.) Sufficient but not complete
6. Which one of the following is a true statement if g is any continuous function?
 - (a.) If T is consistent for θ then $g(T)$ is consistent for θ .
 - (b.) If T is unbiased for θ then $g(T)$ is unbiased for θ .
 - (c.) If T is sufficient for θ then $g(T)$ is sufficient for θ .
 - (d.) All the above three options are false.
7. Three random observations X_1, X_2, X_3 are made on a Poisson distribution with parameter λ .
Let $g(\lambda) = e^{-\lambda}$. Then the Cramer Rao lower bound for the variance of unbiased estimators of $g(\lambda)$ is given by
 - (a.) $\frac{\lambda e^{-2\lambda}}{3}$
 - (b.) $e^{-1/3}$
 - (c.) $\lambda^2 e^{-\lambda}$
 - (d.) $e^{-3\lambda}$
8. Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with density

$$f(x; \theta) = \begin{cases} \theta e^{-\theta x}; & x > 0; \theta > 0 \\ 0; & \text{otherwise} \end{cases}$$
 The moment estimator of θ is
 - (a.) $\sum X_i / n$
 - (b.) Sample median
 - (c.) $n / \sum X_i$
 - (d.) $(\sum X_i)^{\frac{1}{n}}$

9. Let $X_1, X_2, \dots, X_n$ be a random sample of observations from a population with mean θ and finite variance. For estimating θ , the statistic $T_n = \frac{2 \sum_{i=1}^n iX_i}{n(n+1)}$ is
- Biased and consistent
 - Unbiased but not consistent
 - Unbiased and consistent
 - Biased and not consistent
10. Which one of the following statements is true?
- In sufficiency we reduce original random variables to a few statistics for the purpose of drawing inference about parameters.
 - Sufficiency amounts to replacing $X_1, X_2, \dots, X_n$ by a few statistics $T_1, T_2, \dots, T_k$ in the process of discarding the information which is irrelevant to θ .
 - $T_1, T_2, \dots, T_k$ will be called sufficient to θ if the conditional distribution of $T_1, T_2, \dots, T_k$ given $X_1, X_2, \dots, X_n$ is independent of θ .
 - A statistics is said to be sufficient if and only if the point probability density function of random variables can be expressed as two factors, where one factor depends on parameter(s) and observations through the statistics while the other is independent of parameter(s).
11. Which one of the following statements is true?
- The arithmetic mean of a random sample from some population with a finite mean is unbiased for the population mean.
 - If the population variance σ^2 is finite then the sample variance $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ based on a sample $X_1, X_2, \dots, X_n$ is an unbiased estimate of σ^2 .
 - If the r^{th} population moment exists, then r^{th} sample moment is unbiased for the corresponding population moment.
 - In general, the function of any unbiased estimates is unbiased.
12. Let x_1 and x_2 be two independent observations of a Bernoulli random variable that takes values 1 or 0 with probabilities θ and $(1-\theta)$ respectively. If $\theta \in \left[\frac{1}{3}, \frac{2}{3}\right]$, which of the following is not maximum likelihood estimate of θ is?
- $\frac{x_1 + x_2}{2}$
 - $\frac{2 + x_1 + x_2}{6}$
 - $\frac{x_1 + 2x_2}{6}$
 - $\frac{3 + 2x_1 + 2x_2}{6}$
13. If a distribution possesses the monotone likelihood ratio property, then which of the following is incorrect?
- for every randomized test, there exists an equally good non randomized test
 - there exists a uniformly most powerful test for one-sided alternatives for testing composite hypotheses
 - the likelihood ratio test statistic is distributed normally
 - expectation of the likelihood ratio test statistic is equal to zero.
14. Let $X_1, X_2, \dots, X_n$ be a random sample from $U(0, \theta)$. Which of the following is not an unbiased estimator of θ ?
- $\sum x_i / n$
 - $\left(\frac{n+1}{n}\right) \max(X_1, \dots, X_n)$
 - $\left(\frac{n+1}{n}\right) \min(X_1, \dots, X_n)$
 - $(\text{Max}(x_1, \dots, x_n) - \text{Min}(x_1, \dots, x_n))$
15. Which one of the following statements is false?
- For n pairs of observations the maximum possible total score in Kendall's tau is $n(n-1)/2$.
 - Kendall's tau can not be obtained from Karl Pearson's correlation coefficient.
 - Kendall's tau can not be obtained in the case of ties.
 - Kendall's tau can not be adapted to the theory of sampling as Spearman's rank correlation coefficient can be.

16. Which one of the following statements is true?
- In sufficiency we reduce original random variables to a few statistics for the purpose of drawing inference about parameters.
 - Sufficiency amounts to replacing $X_1, X_2, \dots, X_n$ by a few statistics $T_1, T_2, \dots, T_k$ in the process of discarding the information which is irrelevant to θ .
 - $T_1, T_2, \dots, T_k$ will be called sufficient to θ if the conditional distribution of $T_1, T_2, \dots, T_k$ given $X_1, X_2, \dots, X_n$ is independent of θ .
 - A statistics is said to be sufficient if and only if the point probability density function of random variables can be expressed as two factors, where one factor depends on parameter(s) and observations through the statistics while the other is independent of parameter(s).
17. Which one of the following statements is true?
- The arithmetic mean of a random sample from some population with a finite mean is unbiased for the population mean.
 - If the population variance σ^2 is finite then the sample variance $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ based on a sample $X_1, X_2, \dots, X_n$ is an unbiased estimate of σ^2 .
 - If the r^{th} population moment exists, then r^{th} sample moment is unbiased for the corresponding population moment.
 - In general, the function of any unbiased estimates is unbiased.
18. Let x_1 and x_2 be two independent observations of a Bernoulli random variable that takes values 1 or 0 with probabilities θ and $(1-\theta)$ respectively. If $\theta \in \left[\frac{1}{3}, \frac{2}{3}\right]$, which of the following is not maximum likelihood estimate of θ is?
- $\frac{x_1 + x_2}{2}$
 - $\frac{2 + x_1 + x_2}{6}$
 - $\frac{x_1 + 2x_2}{6}$
 - $\frac{3 + 2x_1 + 2x_2}{6}$
19. $X_1, \dots, X_n$ is a random sample from the $U(a-\theta, b+\theta)$, $a < b$, $\theta \geq 0$. Let $X_{(1)} = \min(X_1, \dots, X_n)$, $X_{(n)} = \max(X_1, \dots, X_n)$ and $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, the joint sufficient statistic for (a, θ, b) is
- $(-X_{(1)}, X_{(n)})$
 - $(X_{(1)}, \bar{X}, X_{(n)})$
 - $(X_{(1)} + X_{(n)})$
 - $(X_1, \dots, X_n)$
20. If the characteristic function of i.i.d. random variables X_1 and X_2 is $\phi(t)$ where $\phi: \mathbb{R} \rightarrow \mathbb{R}$, then the characteristic function of $X_1 - X_2$ is
- $\phi(t)$
 - $\phi^2(t)$
 - $-\phi(t)$
 - $\phi\left(\frac{1}{t}\right)$
21. If $X_1, \dots, X_n$ is a random sample of size 'n' from Cauchy $(1, \theta)$, then a consistent estimator for θ is
- Sample Mean
 - Sample variance
 - Sample Median
 - Maximum of $(X_1, \dots, X_n)$
22. If a distribution possesses the monotone likelihood ratio property, then which of the following is incorrect?
- for every randomized test, there exists an equally good non randomized test
 - there exists a uniformly most powerful test for one-sided alternatives for testing composite hypotheses
 - the likelihood ratio test statistic is distributed normally
 - expectation of the likelihood ratio test statistic is equal to zero.
23. If W_1 and W_2 are the most powerful critical regions of size α_1 and α_2 then $\alpha_1 < \alpha_2$ implies
- $W_1 \supset W_2$
 - $W_1 \subset W_2$
 - $W_1 \cap W_2 = \phi$
 - No comparison of W_1 and W_2 is possible

24. For which one of the following test problem UMP test exist
- $H: \mu = \mu_0$ vs $K: \mu \neq \mu_0$ in the case of $N(\mu, 1)$
 - $H: \theta = \theta_0$ vs $K: \theta \neq \theta_0$ in the case Cauchy $(\theta, 1)$
 - $H: \theta = \theta_0$ vs $K: \theta \neq \theta_0$ in the case of Poisson with parameter θ
 - $H: \theta = \theta_0$ vs $K: \theta \neq \theta_0$ in the case $U(0, \theta)$

25. Which one of the following is a false statement?

- A minimal sufficient statistic need not be unique.
- A complete sufficient statistic is minimal.
- A minimal sufficient statistic is a function of all the sufficient statistics.
- A minimal sufficient statistic is complete.

26. If $\hat{\theta}$ is the MLE of θ , which one of the following is false?

- $\hat{\theta}$ is consistent for θ and asymptotically unbiased.
- If g is differentiable function of θ then MLE of $g(\theta)$ is $g(\hat{\theta})$.
- Asymptotically $\hat{\theta}$ is the UMVUE of θ .
- $\hat{\theta}$ is sufficient for θ .

27. A random sample $X_1, X_2, \dots, X_n$ is drawn from a distribution with density function

$$f(x, \theta) = \theta(1-x)^{\theta-1}; 0 < x < 1$$

$$\theta > 0$$

a sufficient statistics for θ is

- $\sum X_i$
- Sample median
- $\sum \log(1 - X_i)$
- $\sum X_i^2$

28. Let $(X_1, X_2, \dots, X_n)$ be a random sample of observations with mean μ and finite variance. Then for estimating μ , the statistic

$$T_n = 2 \sum_{i=1}^n i X_i / n(n+1) \text{ is}$$

- unbiased and consistent
- biased and consistent
- unbiased but not consistent
- biased and not consistent

29. X_1, X_2, X_3 are independent observations from a normal population with mean θ and variance unity.

Two statistics T_1 and T_2 are defined as $T_1 = \frac{X_1 + X_2 + 2X_3}{4}$ and $T_2 = \frac{X_1 + X_2 + X_3}{3}$.

The efficiency of T_1 relative to T_2 is given by

- $\frac{8}{9}$
- $\frac{3}{4}$
- $\frac{9}{8}$
- $\frac{4}{3}$

30. Let ϕ be a test for testing $H: \theta \in \theta_0$ against $K: \theta \in \theta_1$. Then the size of the test of

31. **Assertion (A):** We speak of consistency of a sequence of estimates rather than one point estimate.

Reason (R): Consistency is essentially a large sample property.

Choose the correct answer.

- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is not the correct explanation for A.
- A is true but R is false.
- A is false but R is true.

32. Match List 1 and List 2 and select the correct answer using the codes given below the list.

List 1

List 2

- | | |
|-----------------------|--|
| 1 Randomized Test | A Probability of ejecting the null hypothesis when the alternative hypothesis is true. |
| 2 Non Randomized Test | B $0 \leq \phi(x) \leq 1$ |
| 3 Level of the test | C Probability of rejecting the Null hypothesis when it is true. |
| 4 Power of the test | D $\phi(x) = 0$ or 1 |
| 5 Size of the test | |

The correct match is

- | | A. | B. | C. | D. |
|------|----|----|----|----|
| (a.) | 4 | 1 | 5 | 2 |
| (b.) | 4 | 2 | 1 | 3 |
| (c.) | 5 | 3 | 2 | 1 |
| (d.) | 2 | 1 | 3 | 5 |

ASSIGNMENT SHEET-4

- A system has three components and the system works if at least two of the three components work. The life times of the components are independent and identically distributed exponential random variables with mean 1. If X denotes the life time of the system, then $E(X)$ is
 - 1
 - $\frac{2}{3}$
 - $\frac{5}{6}$
 - $\frac{1}{2}$
- Let $\underline{X}$ be a p dimensional random vector which follows $N(0, I_p)$ distribution and let A be a real symmetric matrix. Which of the following is true?
 - $\underline{X}^T A \underline{X}$ has a chi-square distribution if $A^2 = A$ but the converse is not true
 - If $\underline{X}^T A \underline{X}$ has a chi-square distribution then the degrees of freedom is equal to p
 - If $\underline{X}^T A \underline{X}$ has a chi-square distribution then the characteristics roots of A are either 0 or 1
 - If $\underline{X}^T A \underline{X}$ has a chi-square distribution then A is necessarily positive definite.
- For a random variable X with probability density function

$$f(x) = (1 + \alpha x^{\alpha-1}) e^{-(x+x^\alpha)}, x > 0, \alpha > 0,$$
 the hazard function can be
 - Constant for some α
 - An increasing function for some α
 - A decreasing function for some α
 - A bathtub-shaped function for some α
- The hazard rates of two life time variables T_1 and T_2 with respective c.d.f.s $F_1(t)$ and $F_2(t)$ and p.d.f.s $f_1(t)$ and $f_2(t)$, are $h_1(t) = 3t^2$ and $h_2(t) = 4t^2, t > 0$ respectively. Then
 - $F_1(t) \leq F_2(t)$ for all $t > 0$
 - $F_1(t) < F_2(t)$ for all $t > 1$
 - $E(T_1) < E(T_2)$
 - $f_1(t) < f_2(t)$ for all $t > 0$
- The failure rate of a parallel system of two components, where the component lifetimes are independent and have the exponential distribution with mean 2, is
 - A constant
 - A monotone and bounded function
 - A monotone and unbounded function
 - A non-monotone function
- Let X_1, X_2 and X_3 be independent with $X_1 \sim N(1, 1), X_2 \sim N(-1, 1)$ and $X_3 \sim N(0, 1)$. Let

$$q_1 = \frac{X_1^2 + X_2^2 + 2X_3^2 + 2X_1X_2}{2},$$

$$q_2 = \frac{X_1^2 + X_2^2 + 2X_1X_2}{2}.$$
 Then which of the following statements are always true?
 - q_1 has a central chi-square distribution
 - q_2 has a central chi-square distribution
 - $q_1 + q_2$ has a central chi-square distribution
 - q_1 and q_2 are independent.
- Consider a series system with two independent components. Let the component lifespan have an exponential distribution with density

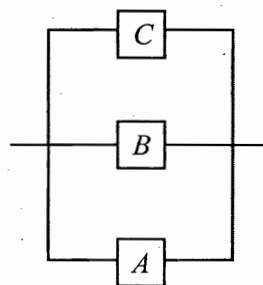
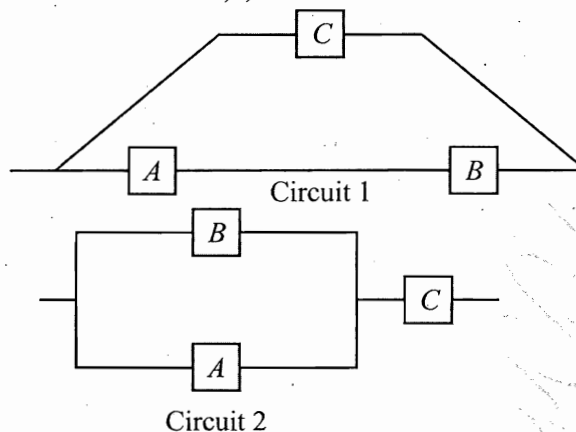
$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & \lambda > 0, x > 0 \\ 0, & \text{otherwise} \end{cases}$$
 If n observation $X_1, X_2, \dots, X_n$ on lifespan of this component are available and $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, then the maximum likelihood estimator of the reliability of the system is given by
 - $(1 - e^{-t/\bar{X}})^2$
 - $1 - (1 - e^{-t/\bar{X}})^2$
 - $e^{-2t/\bar{X}}$
 - $1 - e^{-2t/\bar{X}}$

8. Suppose X is a random variable with following pdf

$$f(x) = \begin{cases} pe^{-x} + 2(1-p)e^{-2x}; x > 0, \\ 0, \text{ otherwise,} \end{cases}$$

and $0 \leq p \leq 1$. Then the hazard function of X is a

- (a.) Constant function for $p = 0$ and $p = 1$
(b.) Constant function for all $0 \leq p \leq 1$
(c.) decreasing function for $0 < p < 1$
(d.) non-monotone function for all $0 < p < 1$
9. Three types of components are used in electrical circuits 1,2,3 as shown below



Circuit 3

Suppose that each of the three components fail with probability p and independently of each other. Let $q_i = \text{Prob}(\text{Circuit } i \text{ does not fail})$; $i=1,2,3$. For $0 < p < 1$, We have

- (a.) $q_3 > q_1$.
(b.) $q_1 > q_2$.
(c.) $q_2 > q_1$.
(d.) $q_2 > q_3$.